

Appendix A

Mathematical Analysis

A.1 Notational conventions

A.1.1 General notational conventions

In this appendix, we elaborate on the mathematical details behind our theory. Below are the notational conventions we adopt:

- *Lowercase letters* (Arabic or Greek) refer to elements of sets (e.g. $i \in N$) or to functions (e.g. $\sigma: N \rightarrow \mathbb{R}$, where $\sigma(i) = x$ means σ maps $i \in N$ to $x \in \mathbb{R}$).
- *Capital letters* (e.g. G, N, S) refer to sets or set-valued correspondences.
- *Calligraphic letters* (e.g. $\mathcal{X}$) refer to sets of sets.
- *Curly braces* $\{\dots\}$ are used to define sets, as in $X \equiv \{x \mid x \text{ is an even integer}\}$. If X is a set, then $\mathcal{X} \equiv 2^X$ denotes its power set (the set of all subsets of X).
- *Profiles* are ordered lists of mathematical objects, such as

$$x \equiv (x_1, \dots, x_n) \in X^n.$$

- *Set subtraction* is denoted $X \setminus Y$, meaning X with the elements of Y removed.
- *Finite sets*: We assume all sets are finite unless stated otherwise.
- If x is an objective quantity in the world, then “ $\hat{x}$ ” indicates the agent’s subjective perception of that object.

- *Terminal-time notation is suppressed in the tables.* When needed, a subscript T indicates terminal objects (e.g. S_T terminal states, H_T terminal histories, h_T a terminal history). We omit these from the notation tables because they are the terminal-time counterparts of S_t , H_t , and h_t .
- *Probability notation.* $\Delta(X)$ denotes the set of probability distributions on a finite set X (deterministic choices are special cases). We omit $\Delta(\cdot)$ from the tables for brevity.
- *Nature's feasible actions.* We allow Nature's feasible action set to depend on the agent's move, writing $A_N(s, a_i)$. When Nature's feasibility does *not* depend on a_i , interpret $A_N(s, a_i) \equiv A_N(s)$ for all $a_i \in A_i(s)$, and drop the second argument when convenient.
- *Induced path distributions.* We write p_σ for the distribution on terminal histories induced by a behavior profile $\sigma = (\sigma_i, \sigma_N)$. We write $p_{(\sigma_i, \sigma_N)}$ only when we need to display the components explicitly; we omit the expanded form from the tables.
- *Prefix-event notation.* If $p \in \Delta(H_T)$ and $h_t \in H_t$, then

$$p(h_t) \equiv p(\{ h_T \in H_T \mid h_t \text{ is a prefix of } h_T \}).$$

Thus, statements such as $p(h_t) > 0$ refer to the cylinder event determined by the paths passing through nonterminal history h_t .

A.2 Basic elements of the theory

States and timeline We adopt a finite horizon $T < \infty$ and assume Time unfolds in discrete periods $t = 0, 1, \dots, T$. A *state of the world* (or simply *state*) at time t is denoted s_t . Then, S_t denotes the set of all *possible* states that could occur at time t . Thus,

$$S \equiv \bigcup_{t=0}^T S_t$$

is the entire state space—the set of all states that could ever possibly occur (over times $0, 1, 2, \dots, T$). *Terminal states* are all those that could occur at time T ; so S_T is the set of *terminal* states. The entire state space S (together with the partition $\{S_t\}_{t=0}^T$) is a *primitive* of our setup. The *initial* or *root* state, s_0 , is realized at the beginning of time.

Agents and action sequences: In what follows, we focus on the capabilities, behaviors, and performances of a sequence of agents, each faced with a dynamic decision problem (interpreted as choosing the actions that will make for an optimal life). The consequences of an agent’s decisions are determined by a “Nature” agent. The Nature agent represents everything that happens during a period that, in conjunction with the action of the focal agent, determines the state of the world in the next period. Nature includes the actions of other agents and events arising from natural forces. Within a period, the sequence of actions is as follows:

1. At $t = 0$, *Nature* takes a solitary action (the focal agent does not move yet) that determines the successor state $t = 1$. This accommodates settings in which the focal agent lands in one of multiple worlds or scenarios. It allows for situations in which the agent is uncertain or unaware of foundational aspects of herself or her environment.
2. From period 1 onward:
 - The *focal agent* i takes an action,
 - Then *Nature* (an aggregate of all exogenous factors) takes an action that, as described above, represents all the happenings that, jointly with the Agent’s action, determine the state in the next period.
 - Then a new state $s_{t+1} \in S_{t+1}$ is realized.

Terminal-state convention. Once a terminal state $s_T \in S_T$ is reached, the dynamic process ends immediately. Thus no further agent move, no Nature move, and no successor state are defined after s_T . In essence, the agent is assumed to be dead on or before period T

Agent actions. At any state s_t for $T > t \geq 1$ in which the focal agent i is alive, the set of distinct *potential actions by the agent* is $A_i(s_t)$. For notational uniformity we also set $A_i(s_0) = \{a_\emptyset\}$, interpreting $a_\emptyset$ as a null action (the agent makes no substantive move at $t = 0$). These actions do not depend on Nature’s move in the *same* period, because the agent moves before Nature for $t \geq 1$. If the agent is dead in s_t , then $A_i(s_t) = \{a_\emptyset\}$. Once the agent is dead at some time t , the agent remains dead in all successor states. Beyond s_0 and provided the agent is alive, she has some non-trivial decision to make: i.e., $|A_i(s_t)| \geq 2$. Since every terminal state ends the action, $A_i(s_T) = \{a_\emptyset\}$.

Nature’s actions. Nature is typically allowed to “observe” the agents action and react to it. Think of Nature as determining the consequences to the agent of her actions. Thus, we write $A_N(s_t, a_{i,t})$ to denote the set of distinct *potential actions available to Nature* when the agent chooses action $a_{i,t}$ from state s_t . At $t = 0$, the agent’s null action is $a_{i,0} = a_\emptyset$, so Nature’s initial move is $a_{N,0} \in A_N(s_0, a_\emptyset)$. At a terminal state s_T , $A_N(s_T) = \{a_\emptyset\}$.

Feasible action profiles. In any state $s_t \in S_t$, $t < T$, the combined period- t move is the *period t action profile*

$$a_t \equiv (a_{i,t}, a_{N,t}),$$

where $a_{i,t} \in A_i(s_t)$ and $a_{N,t} \in A_N(s_t, a_{i,t})$. Let

$$A(s_t) \equiv \{(a_{i,t}, a_{N,t}) : a_{i,t} \in A_i(s_t), a_{N,t} \in A_N(s_t, a_{i,t})\}$$

be the set of all *feasible* action profiles in s_t .¹ After s_0 , Nature’s actions represent the consequences of the Agent’s action (in addition to relevant world events that are not caused by the agent). Then, $A \equiv \bigcup_{s_t \in S \setminus S_T} A(s_t)$ is the set of all action profiles.

State transitions: a tree structure Time starts in state s_0 and flows forward in a *tree* structure. Specifically, assume the agent’s reality is described by a tree

¹When Nature’s feasible set does *not* depend on the agent’s chosen action, we drop the second argument and write $A_N(s_t)$ as shorthand for $A_N(s_t, a_{i,t})$.

$\Gamma \equiv (S, E)$ with node set S and edge set E . Since Γ is a tree, for each period $0 \leq t < T$, there exists a *transition map*

$$f_t : \{(s_t, a_t) \mid s_t \in S_t, a_t \in A(s_t)\} \longrightarrow S_{t+1}$$

such that $f_t(s_t, a_t) = s_{t+1}$, where s_{t+1} is the direct descendant of s_t reached by taking action profile a_t . We define the edge set E by

$$(s_t, s_{t+1}) \in E \iff \exists a_t \in A(s_t) \text{ such that } f_t(s_t, a_t) = s_{t+1}.$$

The action profile a_t is the label on the edge from s_t to s_{t+1} . The tree structure of Γ guarantees no two distinct (s_t, a_t) map to the same s_{t+1} . Similarly, every $s_{t+1} \in S_{t+1}$ arises from some feasible predecessor pair (s_t, a_t) . Thus, along any realized sequence of states $(s_0, \dots, s_T)$ a corresponding sequence of action profiles $(a_0, \dots, a_{T-1})$ is determined by the equalities $f_t(s_t, a_t) = s_{t+1}$.

Histories Paths through the state tree are called *histories*. Formally, for each state $s_t \in S_t$, define the *history at s_t* as

$$h(s_t) \equiv (s_0, s_1, \dots, s_t),$$

the unique sequence of states leading from the initial state s_0 to s_t . We use h_t to denote an arbitrary period- t history (i.e. when there is no need to pin the history to a particular state). The special case $h_0 \equiv (s_0)$ is referred to as the *null history*.

Passing through an action. Suppose $h(s_k) = (s_0, \dots, s_t, s_{t+1}, \dots, s_k)$ is a path in the tree. If a_t is the edge from s_t to s_{t+1} along this path, we say that $h(s_k)$ *passes through a_t* . Since $a_t = (a_i, a_N)$, we may also say it passes through either a_i or a_N individually.²

Terminal histories. A *terminal history* is any history $h(s_T)$ ending in a terminal state $s_T \in S_T$. Let H_T be the set of all such terminal histories, with an arbitrary element denoted h_T . If at time t the *actual* state is s_t , we call $h(s_t)$ the *actualized history* up to time t . For $t < T$, the partial path $h(s_t)$ is the *period- t prefix* of all longer paths (terminal or otherwise) that extend $h(s_t)$. Hence, the *potential histories* from $h(s_t)$ are precisely all continuations of $h(s_t)$ in the subtree rooted at s_t and ending in a terminal state.

²We will use this idea later, for instance, to track when the focal agent's action or Nature's action appears along some path.

Sets of histories. Let H_t denote the set of all period- t histories. Then

$$H \equiv \bigcup_{t=0}^T H_t$$

is the set of all possible histories at every time period. Note that there is a *bijection* between H and S : each state s_t corresponds to exactly one and only one history $h(s_t)$.³ Consequently, any function or correspondence defined on S can equivalently be viewed as being defined on H . For example, if it is understood from the context that $h_t = h(s_t)$, we may write $A_i(h_t)$ instead of $A_i(s_t)$ without ambiguity.

Subtrees For every $s \in S \setminus S_T$, we write $\Gamma_s = (M_s, E_s)$ to denote the full subtree in Γ rooted at s . Here, there is nothing special about “ M ” beyond avoiding awkward notation like $S_s \subset S$.

Behavior profiles and feasible terminal paths Finally, we require a way to talk about what the agent and Nature do, from the God’s-eye perspective of looking down on the world and observing the action. In the following discussion, we adopt the notion of an agent *behavior* from the concept of a “strategy” in game theory. Formally, an agent behavior indicates which feasible action an agent would choose at any state in the state tree should the agent happen to land there.

Agent behavior. Focal agent i ’s *behavior* is a state-contingent rule σ_i such that, for every state $s \in S \setminus S_T$,

$$\sigma_i(\cdot \mid s) \in \Delta(A_i(s)).$$

Thus, $\sigma_i(a_i \mid s)$ is the probability that agent i chooses $a_i \in A_i(s)$ upon landing in s . Note that this structure accommodates stochastic variation in the agent’s action choices (e.g. due to the introduction of randomness, such as flipping a coin).

The agent’s action choices are considered to be freely and intentionally made. Some of the agents that we explore have the super-human ability to choose a σ_i —a state-contingent plan for every conceivable state—at the beginning of time. More realistic agents grope their way through time with limited information, making choices as best they can.

³Thus, each state s_t not only encodes the local features of the world at time t , but implicitly identifies the unique sequence of preceding states $(s_0, \dots, s_{t-1})$.

Nature's behavior. Similarly, Nature's behavior is a state-contingent rule σ_N such that, for every $s \in S \setminus S_T$ where (s, a_i) is a feasible pair (i.e., $a_i \in A_i(s)$),

$$\sigma_N(\cdot \mid s, a_i) \in \Delta(A_N(s, a_i)).$$

Thus, $\sigma_N(a_N \mid s, a_i)$ is the probability that Nature chooses $a_N \in A_N(s, a_i)$ after the agent chooses a_i at s . Deterministic choices are special cases; e.g. $\sigma_N(a_N \mid s, a_i) = 1$ for some $a_N \in A_N(s, a_i)$. Nature's behavior is a primitive of the world that is set at the beginning of time.

Let Σ_i and Σ_N be the sets of all agent and Nature behaviors, respectively. Then, a *behavior profile* σ is then a pair $(\sigma_i, \sigma_N) \in \Sigma$ where $\Sigma \equiv \Sigma_i \times \Sigma_N$ is the set of all such profiles. The idea is that a behavior profile σ describes what each agent does at each state, allowing for the possibility of randomized choice.

It can be shown that each behavior profile σ implies a probability distribution on the set of terminal nodes. Therefore, for each $\sigma \in \Sigma$, let $p_\sigma \in \Delta(H_T)$ be the probability distribution on H_T induced by σ . If the behavior is deterministic, then σ identifies a unique path through the tree. When we hold σ_N fixed and vary σ_i (or vice-versa), we write $p_{(\sigma_i, \sigma_N)}$ to keep the dependence transparent.

Finally, notice that unless σ_N is strictly positive, some paths in H_T are not reachable no matter what behavior Agent i adopts.⁴ We refer to a terminal history $h_T \in H_T$ as *available given σ_N* if there exists a behavior $\sigma_i \in \Sigma_i$ such that $p_{(\sigma_i, \sigma_N)}(h_T) > 0$.

A.3 The A family of agents

A.3.1 Agent A: unlimited knowledge.

We assume **Agent A** has *unlimited deliberative capacity*. This means two things. First, she is fully aware: she knows, at every date and in every state, what actions are available and what consequences can follow from each action, including Nature's state-contingent actions and the probabilities with which they occur. Second, she faces no within-period cognitive bottleneck: when choosing an action, she can keep the entire relevant future in view at once and fully work through its implications before acting. Later, we will model deliberation as capacity-limited—where the agent can examine only part of the future and can actively hold only a limited amount of

⁴Nature's behavior σ_N is *strictly positive* if, for all $s \in S \setminus S_T$ and $a_i \in A_i(s)$, we have $\sigma_N(a_N \mid s, a_i) > 0$ for all $a_N \in A_N(s, a_i)$.

information in mind at decision time. Agent A is the benchmark case in which those limits never bind.

Thus Agent A is an ideal: given Nature's behavior σ_N , she can foresee the consequences of any feasible behavior σ_i and therefore the induced distribution over terminal histories. What remains unspecified is what she *wants*: without a preference criterion, unlimited deliberation alone does not determine a choice.

A.3.2 Agent $A+$: Desires and Deliberation.

To make a *meaningful choice* among feasible terminal histories, the omniscient agent must have some basis for preferring one path over another. We capture this via a *desire* (or preference) relation $\preccurlyeq$ on H_T . Given two terminal paths $h_T, h'_T \in H_T$,

$$h_T \preccurlyeq h'_T \quad \text{means that the agent (weakly) prefers } h'_T \text{ to } h_T.$$

Additionally, we impose two standard conditions:

1. *Completeness*: Every pair of terminal histories can be compared.
2. *Transitivity*: For any h_T, h'_T, h''_T , if $h_T \preccurlyeq h'_T$ and $h'_T \preccurlyeq h''_T$, then $h_T \preccurlyeq h''_T$.

Given our assumption that H_T is finite, these two conditions ensure there exists a real-valued *desire function*

$$d : H_T \rightarrow \mathbb{R}$$

representing the agent's preferences (i.e. $h_T \preccurlyeq h'_T$ if and only if $d(h_T) \leq d(h'_T)$). We may thus obtain a convenient numerical scale for $A+$'s *desires* over outcomes.

To evaluate stochastic behavior profiles, we impose one further assumption: whenever a behavior profile induces a non-degenerate lottery over H_T , the agent is *risk neutral* with respect to the desire function d . Equivalently, if $p, p' \in \Delta(H_T)$ are two such induced distributions, the agent weakly prefers p' to p if and only if

$$\sum_{h_T \in H_T} d(h_T) p(h_T) \leq \sum_{h_T \in H_T} d(h_T) p'(h_T).$$

Hence the agent evaluates a behavior profile σ by

$$\mathbb{E}(\sigma) \equiv \sum_{h_T \in H_T} d(h_T) p_\sigma(h_T).$$

For behavior profiles $\sigma, \sigma' \in \Sigma$, define the induced weak preference relation

$$\sigma \preceq \sigma' \iff \mathbb{E}(\sigma) \leq \mathbb{E}(\sigma').$$

We reserve $\preceq$ for the desires on histories and $\preceq$ for the induced weak preference relation on behavior profiles. Then, **Agent A+** is defined as Agent A equipped with a desire function.

Choosing a behavior. Because Agent A+ retains *all* of Agent A’s omniscient knowledge and unlimited deliberative capacity, she can foresee Nature’s stochastic behavior σ_N with clarity, and, thus, for any choice of behavior, $\sigma_i \in \Sigma_i$, she can compute $p_{(\sigma_i, \sigma_N)}$. Let

$$\Sigma_i^*(\sigma_N) \equiv \{\sigma_i \in \Sigma_i \mid \forall \sigma'_i \in \Sigma_i : (\sigma'_i, \sigma_N) \preceq (\sigma_i, \sigma_N)\}$$

denote the set of the agent’s *optimal behaviors given* σ_N where $\preceq$ is the relation on profiles defined above (there may be several behaviors that result in equally good outcomes). Then, Agent A+ chooses an optimal behavior in period $t = 1$ and, given her omniscience and cognitive omnipotence, continues to do so at every state she visits through period $T - 1$. From the outside observer’s perspective, the behavior of Agent A+ is consistent with some $\sigma_i \in \Sigma_i^*(\sigma_N)$.

Implications of unbounded cognitive capacity. Even if the agent can describe her life in terms of intermediate goals, she does not face a staged *planning-under-bounded-deliberation* problem: she can evaluate complete contingent behaviors directly. Even if A+’s journey contains intermediate goals (e.g. “earn a PhD,” “become an academic,” “earn an international research reputation,” etc.), she need not break her behavior into stages: because she is omniscient, she can compare an *entire* lifetime of intermediate goals head-to-head. Relatedly, because her cognitive capacity is unconstrained, *there is no opportunity cost associated with her deliberations* (e.g., thinking through the implications of taking an action versus, say, resting). Indeed, at each state, she costlessly identifies the optimal action (or randomized choice of action) given the subtree rooted at that state and her knowledge of Nature’s future behavior — and she implements it. Given her foresight, nor is there a meaningful incremental learning or adaptive planning challenge for Agent A+; her *entire* life is essentially optimized (up to Nature’s irreducible randomness) from the start.

Shifting desires. This approach can be easily extended to allow the agent’s desires to vary over time. In that case, Agent A+ knows not only what Nature will do, but also what each of her future selves will do. Thus, she would treat her future selves, who may have different desires, as part of her environment. For example, the period-1 agent would take her future, state-contingent desire functions as given and optimize accordingly according to her present desires.

A.4 The B family of agents

A.4.1 Agent B : ignorance

The first step in our creature de-construction sequence is to crimp agent $A+$'s cognitive capacities by removing omniscience. We refer to this as agent B . Agent B 's deficiency relative to Agent $A+$ is that, while she still knows the entirety of the state tree, the content of the states, and her preference ordering over terminal histories, she no longer knows what Nature will do; i.e., she does not know σ_N . She remains unbounded in her computational capacities.

In this situation, inheriting Agent $A+$'s preference ordering over terminal histories is not a capability sufficient to give B the ability make choices because B can no longer select a behavior σ_i that, given σ_N , generates the greatest $\mathbb{E}(\sigma)$. To proceed, B must be equipped with some sort of decision-making algorithm in the face of this ignorance.

A.4.2 Agent $B+$: decision-making under ignorance

In this subsection, $B+$ is a family label rather than a single fully specified agent: each item below defines a distinct admissible non-probabilistic decision rule for an agent with B 's information. The non-probabilistic procedures typically discussed in the decision theory literature for agents who do not have access to σ_N include:

- **Dominance reasoning** Agent $B+$ identifies a collection of state-contingent actions that is superior to all other options, regardless of what Nature does. If such a dominant behavior exists, Agent $B+$ can choose it with confidence that it will yield the best feasible outcome compared to her other behavior choices. Unfortunately, dominant actions are not always available. When no single action consistently outperforms others across all possible moves of Nature, dominance reasoning cannot provide a clear choice.
- **Iterated elimination of dominated actions** Agent $B+$ identifies a collection of state-contingent actions that iteratively eliminates those action choices that are strictly dominated by others. This eliminates actions for which there exists another that provides a better outcome regardless of Nature's actions. By systematically removing these inferior choices, Agent $B+$ can potentially narrow down the choices to the optimal ones. This method may not always lead to a unique action, especially when the state tree is such that the procedure results in multiple sets of action choices that cannot be further reduced. That

is, this procedure may leave Agent $B+$ with a set of non-dominated action sequences but fail to provide a clear criterion for selecting the best one among them.

- **Robust actions reasoning** Agent $B+$ identifies a collection of state-contingent actions that perform “reasonably well” across a wide range of behaviors by Nature. This approach aims for consistency and effectiveness, even if the strategies are not optimal in any single scenario. Robust actions typically trade off potential optimality for consistency.
- **Minimax/maximin reasoning** Agent $B+$ identifies a collection of state-contingent actions that minimizes the maximum possible loss by considering all strategies and, for each, assuming Nature selects a behavior that imposes the worst possible outcome on the agent and then choosing the behavior for which this worst-case scenario is the least bad. Of course, the problem here is that minimax reasoning can be overly conservative, potentially leading to suboptimal decisions when the worst-case scenarios are unlikely to occur. Moreover, such reasoning does not differentiate between choices that perform well on average versus those that perform well only in the worst case.

Maximin reasoning is similar to the minimax approach but focuses on maximizing the minimum payoff. Like the minimax approach, maximin reasoning can be overly cautious, leading to conservative decisions that are not optimal in the “average” case.

None of these well-studied algorithms require any inputs beyond a complete and transitive preference ordering on terminal histories, which this agent has inherited from Agent $A+$. Even so, in the face of ignorance about Nature’s behavior, any of these approaches will generally fall short of identifying the optimal actions (given Nature’s actual behavior) because they either rely on overly conservative assumptions, focus on minimizing losses rather than maximizing gains, or do not fully incorporate the likelihood of different outcomes.

A.4.3 Agent $B++$: credences

Alternatively, we can equip Agent B with the ability to fill in the missing element using her imagination. That is, since B is ignorant of σ_N , we can add the cognitive capacity to formulate a *credence* (or *subjective belief*) about what Nature’s behavior is, which we label $\hat{\sigma}_N$. Given a behavior σ_i and subjective model $\hat{\sigma}_N$, we obtain

a *subjective distribution on terminal histories*, $p_{(\sigma_i, \hat{\sigma}_N)}$. In parallel with Agent $A+$, define

$$\mathbb{E}(\sigma_i, \hat{\sigma}_N) \equiv \sum_{h_T \in H_T} d(h_T) p_{(\sigma_i, \hat{\sigma}_N)}(h_T).$$

With this additional capacity, $B++$ is capable of making decisions using the same mechanics as $A+$, with the exception that $\hat{\sigma}_N$ is substituted for σ_N . When σ_i maximizes $\mathbb{E}(\sigma_i, \hat{\sigma}_N)$, the behavior is referred to as *subjectively rational* to distinguish it from the objectively rational behavior of Agent $A+$.

Notice that the only requirement for $B++$ is that she have a credence. There is no assumption that such credences be “close” to reality in any sense. Therefore, it cannot be said that $B++$ generally does better than $B+$.

One way in which decision theorists impose some order on an agent’s credences is to assume that, given the agent’s choice of behavior σ_i , her subjective beliefs $\hat{\sigma}_N$ are correct along the realized path. Specifically, for every terminal history $h_T \in H_T$ such that

$$p_{(\sigma_i, \sigma_N)}(h_T) > 0,$$

her subjective and objective probabilities coincide:

$$p_{(\sigma_i, \hat{\sigma}_N)}(h_T) = p_{(\sigma_i, \sigma_N)}(h_T).$$

The idea is that, due to a combination of societal knowledge, training, and experience, credences are accurate with respect to events that *actually* occur (indeed, such confidence may be why the agent chose those actions to begin with). Nevertheless, credences about what Nature would have done under counterfactual conditions are unrestricted.

The canonical example is a world in which everyone believes the earth is flat. Under such credences, the optimal sailing routes are those that stay close to shore. These credences are true: the optimal choice of action leads to observed outcomes. The counterfactual scenario is never disconfirmed—the actions are subjectively optimal but objectively suboptimal.

A.4.4 Agent $B+++$: Bayesian learning

Priors. For the Bayesian-learning subsection we adopt an equivalent finite representation of Nature. Let $\bar{\Sigma}_N$ denote the set of all *deterministic* behaviors for Nature, i.e., maps

$$\bar{\sigma}_N : \{(s, a_i) : s \in S \setminus S_T, a_i \in A_i(s)\} \rightarrow A_N(s, a_i)$$

that pick a single Nature action at every nonterminal feasible pair (s, a_i) . Because the tree is finite and each feasible pair can be encountered at most once along any history, any stochastic Nature behavior induces the same distribution on terminal histories as a probability distribution on $\bar{\Sigma}_N$. Therefore, it is possible to formulate a Bayesian agent who learns over $\bar{\Sigma}_N$.

Let $\hat{\sigma}_N(\cdot \mid h_0) \in \Delta(\bar{\Sigma}_N)$ be a strictly positive initial prior. For each history $h_t \in H$, let $\hat{\sigma}_N(\cdot \mid h_t) \in \Delta(\bar{\Sigma}_N)$ denote the posterior obtained by conditioning $\hat{\sigma}_N(\cdot \mid h_0)$ on the subset of terminal paths that pass through h_t .⁵ Thus only the initial prior is primitive; later beliefs are generated from it by Bayesian updating.

Given σ_i and $\hat{\sigma}_N(\cdot \mid h_t)$, define the induced predictive distribution on terminal histories by

$$p_{(\sigma_i, \hat{\sigma}_N(\cdot \mid h_t))}(h_T) := \sum_{\bar{\sigma}_N \in \bar{\Sigma}_N} \hat{\sigma}_N(\bar{\sigma}_N \mid h_t) p_{(\sigma_i, \bar{\sigma}_N)}(h_T), \quad h_T \in H_T.$$

Credence Updating Define an indicator-product for the consistency of a deterministic contingency behavior $\bar{\sigma}_N$ with an actualized history h_t as

$$\iota(\bar{\sigma}_N, h_t) \equiv \prod_{\tau=0}^{t-1} \mathbf{1}[\bar{\sigma}_N(s_\tau, a_{i,\tau}) = a_{N,\tau}],$$

where $(a_{i,\tau}, a_{N,\tau})$ is the unique action profile on the edge from s_τ to $s_{\tau+1}$. Thus, $\iota(\bar{\sigma}_N, h_t) = 1$ exactly when $\bar{\sigma}_N$ matches Nature's realized moves at every period along h_t . Since the agent's realized action probabilities are the same under every $\bar{\sigma}_N$, they cancel from Bayes' rule and therefore do not appear in (A.1).

Whenever $p_{(\sigma_i, \hat{\sigma}_N(\cdot \mid h_0))}(h_t) > 0$, the agent updates her belief about which deterministic contingency plan is in force by applying Bayes' rule:

$$\hat{\sigma}_N(\bar{\sigma}_N \mid h_t) = \frac{\hat{\sigma}_N(\bar{\sigma}_N \mid h_0) \iota(\bar{\sigma}_N, h_t)}{\sum_{\tilde{\sigma}_N \in \bar{\Sigma}_N} \hat{\sigma}_N(\tilde{\sigma}_N \mid h_0) \iota(\tilde{\sigma}_N, h_t)}. \quad (\text{A.1})$$

Because $\hat{\sigma}_N(\cdot \mid h_0)$ is strictly positive, the denominator in (A.1) is positive for every history h_t with $p_{(\sigma_i, \hat{\sigma}_N(\cdot \mid h_0))}(h_t) > 0$, so the posterior $\hat{\sigma}_N(\cdot \mid h_t)$ is well defined on-path.

⁵Technically, the event is $E_{h_t} = \{h_T \in H_T : h_T \text{ passes through } h_t\} \subseteq H_T$, conditioned through the joint law on $\bar{\Sigma}_N \times H_T$ induced by $\hat{\sigma}_N(\cdot \mid h_0)$ and σ_i . Since $\bar{\Sigma}_N$ is deterministic, this reduces to restricting the prior's support to the $\bar{\sigma}_N$ whose moves agree with those recorded along h_t , and renormalizing; the agent's actions enter every $\bar{\sigma}_N$'s likelihood identically and cancel. Hence the posterior is independent of σ_i and, by strict positivity of the prior, is defined at every $h_t \in H$ —including histories that σ_i reaches with zero probability.

Equivalently, $\hat{\sigma}_N(\cdot \mid h_t)$ is the conditional distribution over deterministic contingency plans given the cylinder event determined by h_t . If one allows priors that put zero mass on some plans and the denominator vanishes, we adopt the convention that $\hat{\sigma}_N(\cdot \mid h_t) = \hat{\sigma}_N(\cdot \mid h_0)$.

Choice of Behavior For each action $a_i \in A_i(h_t)$ taken at a nonterminal history $h_t \notin H_T$, let

$$p_{(\sigma_i, \bar{\sigma}_N)}(h_T \mid h_t, a_i)$$

be the probability that terminal history h_T is reached when the agent chooses a_i and then follows her behavior σ_i from h_{t+1} onward, and Nature follows the deterministic behavior $\bar{\sigma}_N$. Then, the expected payoff when choosing action a_i at h_t is given by

$$\sum_{\bar{\sigma}_N \in \bar{\Sigma}_N} \hat{\sigma}_N(\bar{\sigma}_N \mid h_t) \sum_{h_T \in H_T} d(h_T) p_{(\sigma_i, \bar{\sigma}_N)}(h_T \mid h_t, a_i).$$

Rationality requires that, at every history h_t reached with positive probability under $(\sigma_i, \hat{\sigma}_N(\cdot \mid h_0))$, a behavior such that $\sigma_i(a_i \mid h_t) > 0$ implies

$$a_i \in \arg \max_{a'_i \in A_i(h_t)} \sum_{\bar{\sigma}_N \in \bar{\Sigma}_N} \hat{\sigma}_N(\bar{\sigma}_N \mid h_t) \sum_{h_T \in H_T} d(h_T) p_{(\sigma_i, \bar{\sigma}_N)}(h_T \mid h_t, a'_i).$$

Interpretation. In brief, B^{+++} behaves like B^{++} but refines her credences about Nature's behavior by Bayesian conditioning on the realized history. Because the horizon is finite, no asymptotic convergence claim is made here. The point is only that posterior mass is reallocated toward those deterministic behaviors that remain consistent with Nature's observed moves.

A.5 The C family of agents

All versions of Agents A and B can represent the entire state tree Γ at each state. In particular, these agents are not simply following a plan they work out at the beginning of time. Rather, before acting in any reached state, they know all potential future states and they can recompute the implications of alternative behaviors using all the information encoded in the entire subtree rooted at that state. Their deliberations are bound by neither memory nor within-period computation.

In the real world, however, even a single major decision demands that the agent formulate objectives, construct a set of candidate options, and gather and process information sufficient to evaluate them—tasks whose cognitive demands make exhaustive consideration of every consequence and contingency infeasible. Consequently,

many future options and potential states of the world never enter the agent's mind. As such, these are not unlikely alternatives to which the agent assigns negligible probability — rather, they are possibilities that remain entirely unconceived. Decision theorists call this phenomenon *unawareness* — a constraint on cognition distinct from *uncertainty* about the likelihood with which events in the state tree will occur.

We now investigate agents who face deliberation under unawareness: we build three variants $C \rightarrow C+ \rightarrow C^{\text{Plan}}$. Our first agent is one who is aware of only a small portion of the state tree in her immediate future. We then endow her with increasingly effective means of mitigating this limitation: first with the ability to explore portions of the tree downstream from her present state and, second, with the ability to plan.

None of the C agents are explicitly endowed with long term memory. Rather, once an agent has thought through a decision problem (e.g., the agent has researched the pros and cons of pursuing a Ph.D.) and made choices about a future set of state-contingent actions (e.g., choosing a list of candidate schools and making acceptance plans contingent on acceptances), she reduces those choices to a plan (writes them down, enters them into a spreadsheet, or just commits them to memory). She then follows the plan as long as her experiences are consistent with it.⁶ Thus, the plan serves as a limited memory device for the Agent's state-contingent action strategy. However, as we will see, it is not equivalent to recalling all elements of previously discovered information and the analyses conducting on it at the point of plan formation..

First, we introduce a few notational conventions that will let us speak precisely about the agent's *partial* view of any continuation tree (the subtree rooted at the agent's present state):

- The *support* of a history $h_t = (s_0, \dots, s_t)$ is $\text{Supp}(h_t) \equiv \{s_0, \dots, s_t\}$; i.e., $\text{Supp}(h_t)$ converts the component states in the history *profile* h_t into a corresponding *set* of states.
- If $s \in S_t$, then the depth (time index) of s is t and is denoted $\text{Depth}(s)$.
- Let $\text{Par}(s)$ denote the *parent of state* s , $\text{Desc}(s) \subseteq S$ the *set of states that are descendants of* s , and $\text{Chld}(s) \subseteq \text{Desc}(s)$ the *set of direct descendants of* s (i.e., its *children*). Thus, $\text{Par}(s_0) = \emptyset$, and for $s \in S_T$, $\text{Chld}(s) = \text{Desc}(s) = \emptyset$.

⁶In this treatment, her future experiences are always consistent with the plan in the sense that, once the plan is formulated and followed, she never arrives an out-of-plan state. Nevertheless, the relative costs and benefits of investigation and analysis may evolve as the plan unfolds. These changes may induce changes to the plan or even its abandonment.

- We use the hat diacritic $\hat{\cdot}$ to indicate objects that are in the agent's *awareness* and the bar diacritic $\bar{\cdot}$ to indicate objects that are elements of the agent's plan.
- Throughout this section, the agent's *awareness tree at state* $s_t \in S \setminus S_T$ is a partial subtree of Γ rooted at s_t . We represent it by its node set

$$\hat{M}_{s_t} \subseteq \text{Desc}(s_t) \cup \{s_t\},$$

which is assumed to satisfy parent closure: if $s \in \hat{M}_{s_t}$ and $s \neq s_t$, then $\text{Par}(s) \in \hat{M}_{s_t}$. Given Γ , the induced feasible transitions (edges) in the subtree implied by $\hat{M}_{s_t}$ are determined by the objective feasibility correspondences and transition map f : given $\hat{M}_{s_t}$, a profile (a_i, a_N) is an *induced feasible move from* $s \in \hat{M}_{s_t}$ iff $(a_i, a_N) \in A(s)$ and $f_{\text{Depth}(s)}(s, (a_i, a_N)) \in \hat{M}_{s_t}$. Notice that, given the context of Γ , a subtree is implied by $\hat{M}_{s_t}$ since, once $\hat{M}_{s_t}$ is known, so are the edge labels inherited from Γ .

A.5.1 Agent C : extreme unawareness

Agent C has severely restricted awareness of the state tree as she travels through it. Specifically, upon arriving at state s_t , she becomes aware of a one-step-ahead awareness tree represented by the node set

$$\hat{M}_{s_t}^{\text{arr}} \equiv \{s_t\} \cup \text{Chld}(s_t),$$

where $\text{Chld}(s_t)$ is the complete set of direct descendants (children) of s_t .⁷ Agent C at s_t is unaware of all continuation histories beyond the next period. She cannot reason about these states; they are completely out-of-mind.

Inherited from the later B agents, Agent C is equipped with the capacity to form credences regarding Nature's moves: for all $a_i \in A_i(s_t)$ and all $a_N \in A_N(s_t, a_i)$ she forms a credence $\hat{\sigma}_N(\cdot \mid s_t, a_i) \in \Delta(A_N(s_t, a_i))$.⁸

Payoff assessments Unless Agent C is in a second-to-terminal state, her awareness horizon is truncated since she is only aware of one-step-ahead state tree paths. Facing a truncated awareness horizon, her inherited desire function d can no longer be used to assess her available actions. Nevertheless, she must choose an action and,

⁷Earlier versions of this setup allowed for partial awareness of Nature's consequences (more realistic), but this required substantial additional notational bookkeeping with little additional insight regarding our focal issues.

⁸The $C+$ and C^{Plan} agents are assumed to retain this credence-formation capacity for states they discover beyond s_t .

to do so, we must provide her with some method of choice assessment. We proceed to consider settings in which the agent makes coarse assessments that are, nevertheless, *bounded* by reality.

Specifically, define the *objective payoff interval associated with any* $s \in S$ as $[\underline{d}(s), \bar{d}(s)]$ where:

$$\begin{aligned}\underline{d}(s) &\equiv \min_{h_T: \text{Supp}(h_T) \ni s} d(h_T), \\ \bar{d}(s) &\equiv \max_{h_T: \text{Supp}(h_T) \ni s} d(h_T).\end{aligned}$$

Then, for any future state s of which the agent is aware from her present state, she assigns the range $[\underline{d}(s), \bar{d}(s)]$. For example, arriving in s_t , for each $s_{t+1} \in \hat{M}_{s_t}^{\text{arr}}$, she associates the payoff range $[\underline{d}(s_{t+1}), \bar{d}(s_{t+1})]$. If the children are terminal states (i.e., $t + 1 = T$), then she knows the exact payoffs: $\underline{d}(s_{t+1}) = \bar{d}(s_{t+1}) = d(s_{t+1})$.

The agent is unable to elaborate the details associated with the continuation paths emanating beyond a next-period state because they are outside her awareness. The question this poses is the extent to which her decision process is grounded in reality. If that process is completely unmoored from it, then her decisions amount to little more than flipping a coin. On the other hand, knowing the set of payoffs associated with all the terminal histories passing through her present state (as do Agents A and B) is inconsistent with being unaware of what happens beyond her awareness horizon.

The assessment rule defined above imposes what we judge to be a reasonable balance. We assume the agent knows: i) the payoff consequences of arriving at a downstream state of which she is aware will fall within some *range*; and ii) although she remains unaware of the specifics that would allow her to refine her assessment further, she *correctly* assesses bounds of the payoff range passing through s_{t+1} are bounded below by $\underline{d}(s_{t+1})$ and above by $\bar{d}(s_{t+1})$. In other words, her analyses are at least loosely bounded by reality.

For example, in considering whether to pursue a Ph.D. in Philosophy, our agent may know that the range of payoff outcomes is bounded above by getting a job at a top department and below by failing to get any job in academia at all (for both of which we assume she can accurately quantify the associated payoffs). However, she is unaware of the details that lead to these outcomes and, beyond these two extremes, there may be many other payoff possibilities of which she is unaware.

In s_t , considering the next period consequence s_{t+1} and its associated range $[\underline{d}(s_{t+1}), \bar{d}(s_{t+1})]$, the agent makes a *point-estimate*, denoted $\hat{d}(s_{t+1})$, of the specific

value associated with that state according to

$$\hat{d}(s_{t+1}) \equiv (1 - \alpha)\underline{d}(s_{t+1}) + \alpha\bar{d}(s_{t+1}), \quad (\text{A.2})$$

where $\alpha \in [0, 1]$ can be thought of as a measure of *agent optimism* used to convert payoff intervals to point values.⁹ The larger the value of α the greater the weight given to the upper bound.

Then, given these inputs, the agent identifies the set of actions at s_t that maximize her subjective expected payoff:

$$A_i^*(s_t) \equiv \left\{ a_i \in A_i(s_t) \mid a_i \in \arg \max_{a_i \in A_i(s_t)} \sum_{a_N \in A_N(s_t, a_i)} \hat{d}\left(f_t(s_t, (a_i, a_N))\right) \hat{\sigma}_N(a_N \mid s_t, a_i) \right\}. \quad (\text{A.3})$$

If there is more than one optimal action, she chooses $a_i^* \in A_i^*(s_t)$ uniformly.

Summing up, C traverses the state tree and, at each arrival state, she becomes aware of all her feasible actions as well as the potential direct consequences by Nature with which they are associated. She is well-calibrated in terms of the *ranges* of possible continuation payoffs from the follow-on states. She assesses these in accordance with her degree of optimism α and subjective credences $\hat{\sigma}_N$. She then chooses an action, Nature acts, she arrives in a new state in the next period and so on until she reaches a terminal state.

Agent C examples

Example 1 (Repeated Prisoner's Dilemma). *Consider an example with three repetitions of a Prisoner's Dilemma game. Each period the focal agent i chooses C (cooperate) or D (defect), so $a_{i,t} \in \{C, D\}$. Nature observes $a_{i,t}$ and chooses $a_{N,t} \in \{C, D\}$ according to a behavior rule. Payoffs are accumulated each period and paid at the end of the game. For simplicity, the agent does not discount payoffs. The period payoff added to the agent's final total depends only on the action profile $(a_{i,t}, a_{N,t})$ in that period according to: $(D, D) = 1, (C, C) = 4, (D, C) = 5, (C, D) = 0$.*

*At period t , Nature plays C unless the agent defected at $t-1$; once a defection has occurred, Nature plays D forever afterward. This is an environmental regularity, not a strategic opponent. Thus, Nature is in a behavioral **status** of Cooperative, meaning Nature will play C next period, or Punishing, meaning Nature plays D next period and forever thereafter.*

⁹Or, more generally, as a parameter that reflects the agent's preferences over intervals (for whatever reasons).

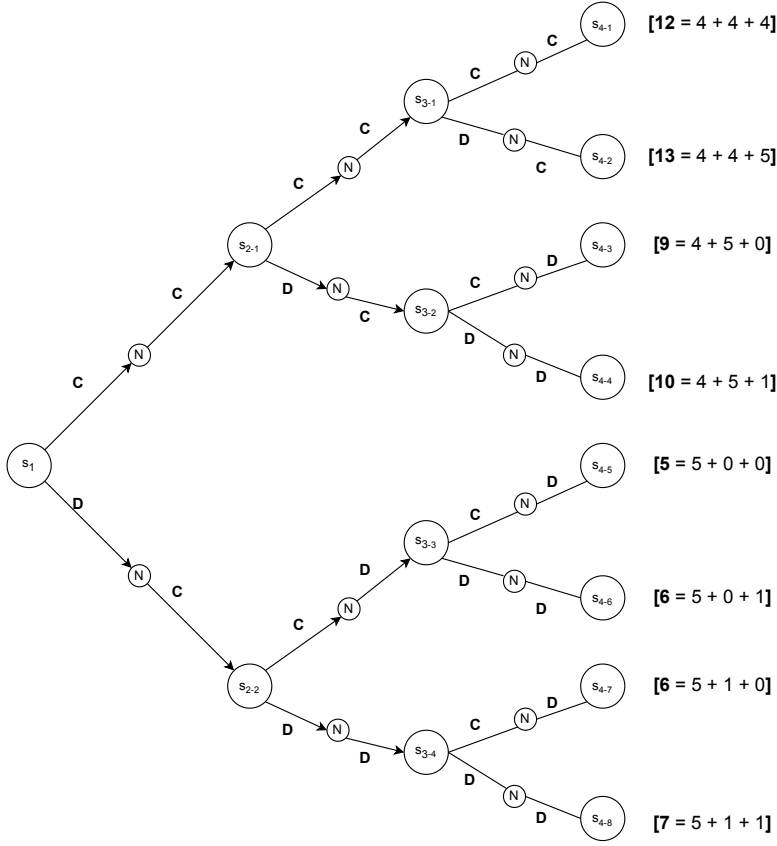


Figure A.1: The state tree with feasible moves given N 's behavior.

Figure A.1 illustrates the state tree, with infeasible paths (given N 's deterministic Grim behavior) pruned from the diagram to avoid clutter. The optimal behavior for the agent is C, C, D . Thus Agent $A+$ adopts C, C, D under the deterministic Grim environment described here.

Upon arrival in s_1 , Agent C is aware of her action choices and the replies of N as in Figure A.2. Faced with this situation, she chooses C and proceeds to node s_{2-1} . The reason for this choice is simply that $\hat{d}(s_{2-1}) > \hat{d}(s_{2-2})$ for every $\alpha \in [0, 1]$, since $[\underline{d}(s_{2-1}), \bar{d}(s_{2-1})] = [9, 13]$ and $[\underline{d}(s_{2-2}), \bar{d}(s_{2-2})] = [5, 7]$. Thus, no matter what the value of $\alpha \in [0, 1]$, the proxy assigned to s_{2-1} exceeds the proxy assigned to s_{2-2} .

Upon arrival in s_{2-1} , Agent C 's awareness is similar to Figure A.2, except that now she is aware of the partial subtree rooted at s_{2-1} and terminating in s_{3-1} and s_{3-2} , with corresponding intervals of $[12, 13]$ and $[9, 10]$, respectively. Using the same

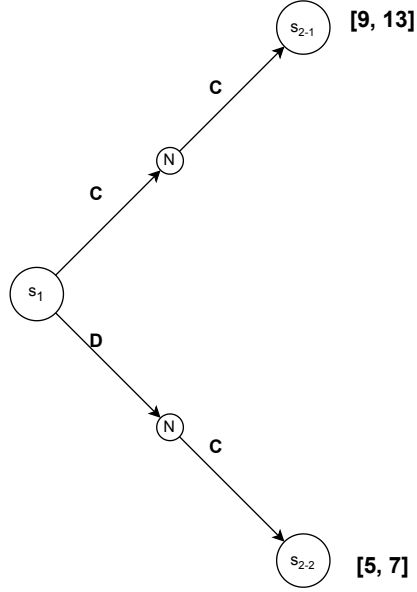


Figure A.2: The tree in $C+$'s long-term memory at the start.

logic as above, she chooses C to move to state s_{3-1} . Here, she sees the final portion of the tree and chooses D . Thus, Agent C has managed to grope her way to optimality!

Example 2 (Failure to optimize). *To show that groping toward optimality is not generally possible, consider the state tree in Figure A.3. Assume $\alpha = 0.5$ and that Nature picks actions with 50% probability. Furthermore, assume the agent is well-calibrated with respect to Nature's behavior.*

It should be easy to see that $\hat{d}(s_{2-1}) = \hat{d}(s_{2-2}) = 5$: both ranges are $[0, 10]$ and $\alpha = 0.5$. Similarly, $\hat{d}(s_{2-3}) = \hat{d}(s_{2-4}) = -50$. Thus, given her awareness in s_1 and her knowledge of Nature's behavior, the subjective expected value of choosing C is 5 and of choosing D is -50 (since Nature chooses her actions with 50% probability). Following this decision (and regardless of which consequence Nature chooses), in $t = 2$, the agent once again opts for C for an expected payoff of 5 (which, at this point, is correct).

Unfortunately, the s_1 choice of C is suboptimal. Instead, she should have chosen D because this would have guaranteed her an expected payoff of 50. Here, we see the problem of computing actions based upon unrefined ranges: upon reaching s_{2-3} (or s_{2-4}), the agent becomes aware of the remaining subtree in its entirety—at which point she chooses C and completely avoids the negative payoff associated with the D choice in that period.

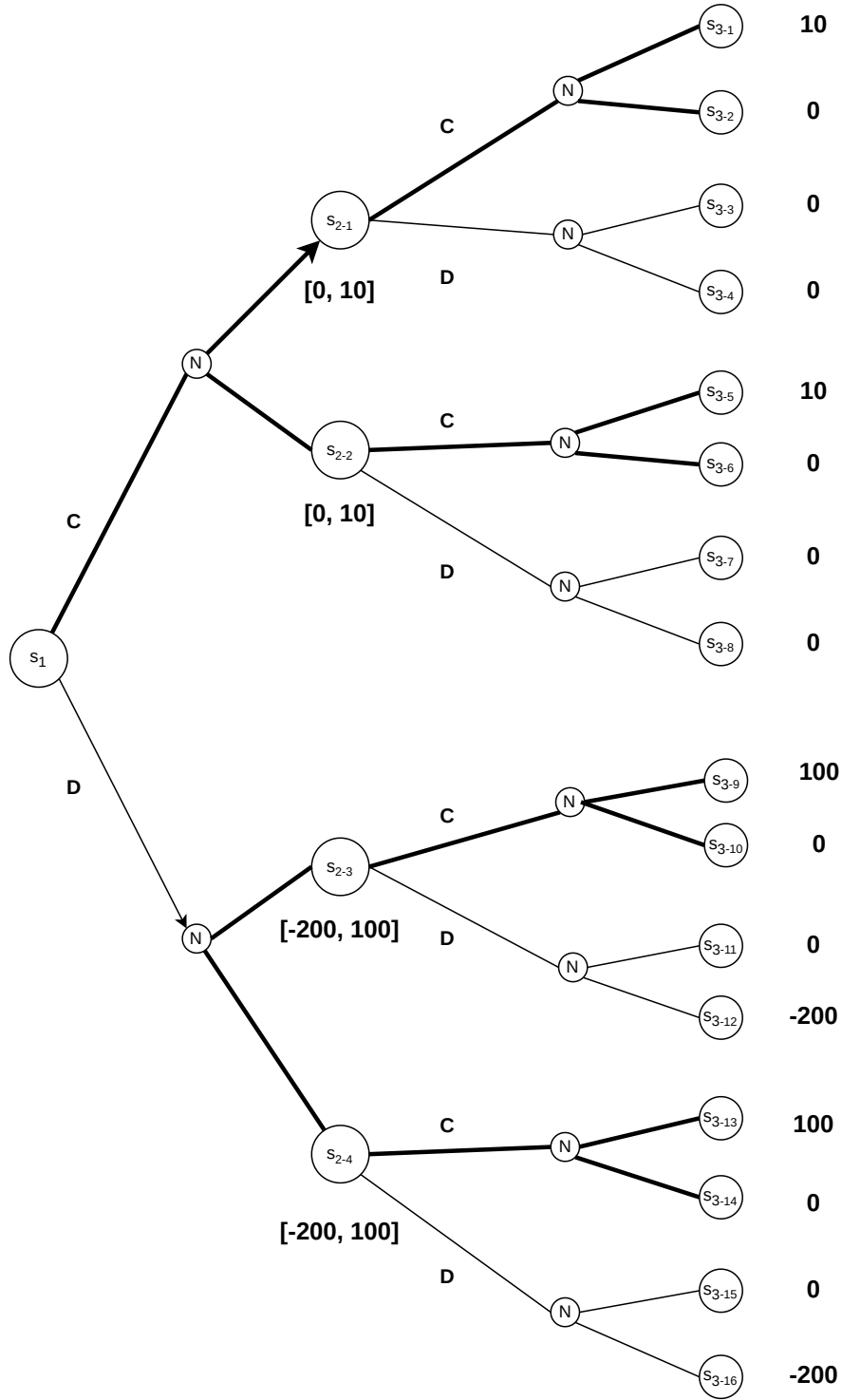


Figure A.3: Abstract tree: $\alpha = 0.5$ and Nature picks actions with 50% probability; bold segments show state-contingent choices by the agent and Nature.

Another potential source of bias is α . In the preceding examples, this preference parameter was either irrelevant or unbiased. Is biased α a problem? The answer depends upon how it is interpreted. If α represents a secondary subjective probability assessment on the part of the agent, then any value that results in an inaccuracy relative to the true expected value of choosing that action may lead to suboptimal choices. However, if α is a preference parameter for evaluating intervals, then it is difficult to say that the choices are suboptimal at the time they are being made because they reflect the agent's true valuation when faced with an interval outcome. Finally, the agent's credences with respect to Nature's behavior may also be wrong. Errors of this kind can also lead to decisions that are objectively suboptimal (as is the case with Agent B+).

A.5.2 Agent $C+$: deliberative exploration

Agent $C+$ extends Agent C by allowing the agent to extend her awareness through intentional investigation and analysis of future events outside of her awareness horizon before acting. In the unawareness literature, a distinction is often made between unawareness of unawareness and awareness of unawareness. As Donald Rumsfeld famously put it,

Reports that say that something hasn't happened are always interesting to me, because as we know, there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say, we know there are some things we do not know. But there are also unknown unknowns — the ones we don't know we don't know. And if one looks throughout the history of our country and other free countries, it is the latter category that tends to be the difficult ones.¹⁰

Our $C+$ agent is endowed with awareness of her unawareness. She knows that there are things “out there” beyond the horizon of her awareness. This provides the motivation to direct her cognitive resources toward investigating interesting branches of the future state tree and bring them into her awareness. We refer to the process of investigation (collecting information about future unknowns) and analysis (evaluating the implications of newly discovered information on the decision at hand) as *deliberation*.

Prior to acting, Agent $C+$ may select a discovered nonterminal frontier state for investigation, reveal its direct successor states, and analyze the implications for her present choice. Since the C -family agents are not assumed to have unlimited cognitive capacities, each investigation-analysis round imposes an opportunity cost: cognitive resources directed toward decision making rather than toward something else, such as reading a book. We treat this as a cost of the current deliberative episode, not as a change in the desire function d , and not as an alteration of terminal path values.

To reflect this, we introduce a *deliberation cost function*

$$\kappa : \mathbb{N}_0 \rightarrow \mathbb{R}_{\geq 0} \cup \{+\infty\},$$

where $\kappa(m)$ is the cognitive cost of completing m deliberative rounds before acting at the agent's present state. Each deliberative round consists of investigating one

¹⁰Wikipedia contributors. (2026, June 10). There are unknown unknowns. In Wikipedia, The Free Encyclopedia. Retrieved 17:07, June 11, 2026, from https://en.wikipedia.org/w/index.php?title=There_are_unknown_unknowns&oldid=1358720428.

expandable frontier state and then analyzing the updated awareness tree in order to recompute the estimated continuation values relevant to the present action choice. The initial assessment of arrival awareness is part of the baseline choice procedure and is not counted as a costly deliberative round. Assume

$$\kappa(0) = 0$$

and that κ is weakly increasing. For $m \in \mathbb{N}_0$, define the *marginal cost of the $(m+1)$ -st deliberative round* by

$$\Delta\kappa(m+1) \equiv \kappa(m+1) - \kappa(m), \quad (\text{A.4})$$

with the convention that $+\infty - c = +\infty$ for finite c .

Agent $C+$'s state- s_t deliberation is organized around the action problem she presently faces. When she arrives at s_t , she must choose among the feasible actions in $A_i(s_t)$. Arrival awareness gives her: i) the feasible actions at s_t ; ii) Nature's feasible responses to those actions; iii) her credences over Nature's responses; and iv) the corresponding successor states. Like Agent C , before any further deliberation, she can compute a subjective expected continuation value for each of her presently feasible actions and, thereby, identify an initial candidate action as the best choice.

The capacity to deliberate is valuable only insofar as it has the potential to change her assessment of the action presently identified as best. It is not undertaken merely because some unresolved state is imprecisely understood. Nor is it necessarily undertaken if extending awareness at a state changes the continuation values among the feasible actions at s_t . Rather, it must have the potential to change the agent's assessment of which action is best.¹¹

The $C+$ deliberation procedure

In what follows we adopt the notational convention to use v to indicate continuation values of *states* and *actions* in various contexts. We need to keep track of both as we proceed through deliberations.

A. Initialization and frontier notation.

1. Set the *loop counter* $k := 0$, where “ $:=$ ” indicates assignment, as in an algorithmic procedure.

¹¹The continuation values of the agent's present feasible actions can change without altering the best choice.

2. When Agent $C+$ arrives at s_t , she (costlessly) discovers the same one-step-ahead awareness tree as Agent C :

$$\hat{M}_{s_t}^{\text{arr}} := \{s_t\} \cup \text{Chld}(s_t).$$

3. Initialize the $k = 0$ discovered *state awareness*, *leaf*, and *expandable leaf* sets:

$$\begin{aligned} M^0 &:= \hat{M}_{s_t}^{\text{arr}}, \\ L^0 &:= \{s \in M^0 \mid \text{Chld}(s) \cap M^0 = \emptyset\}, \\ L_{\text{exp}}^0 &:= \{s \in L^0 \mid s \notin S_T\}, \text{ respectively.} \end{aligned}$$

Terminal leaves are leaves for purposes of assessment, but they are not candidates for expansion (because there is nothing beyond them to reveal).

4. For each discovered leaf $s \in L^0$, set the *estimated continuation value at s* :

$$v^0(s) := \hat{d}(s) = (1 - \alpha)\underline{d}(s) + \alpha\bar{d}(s).$$

If $s \in S_T$, then this assignment is exact because the terminal state determines the corresponding terminal path value.

5. For each feasible action $a_i \in A_i(s_t)$, the *estimated continuation value of choosing that action* is

$$v_{s_t}^0(a_i) := \sum_{a_N \in A_N(s_t, a_i)} v^0(f_t(s_t, (a_i, a_N))) \hat{\sigma}_N(a_N \mid s_t, a_i).$$

The corresponding initial estimated continuation value of the present state is

$$v^0(s_t) := \max_{a_i \in A_i(s_t)} v_{s_t}^0(a_i),$$

and the initial maximizing-action set is

$$A_i^{*,0}(s_t) := \arg \max_{a_i \in A_i(s_t)} v_{s_t}^0(a_i).$$

The value $v_{s_t}^0(a_i)$ is the subjective expected value, according to the initially discovered awareness tree in loop $k = 0$, of choosing a_i at s_t .¹²

¹²Because a discovered state becomes a non-leaf only when its direct successors have been revealed, this expectation is well formed.

Thus, following initialization, the agent who just arrived at s_t has computed continuation values for all the states of which she has just become aware as well as an initial set of continuation values for her feasible actions there.

Agent $C+$ is *endowed with the ability to investigate the frontier states of which she is aware*. In our setup, investigating a state results in the agent obtaining the same information as if she arrived there in reality — that is, she becomes aware of the children of that state. The candidates for investigation are the expandable leaves.

B. Identifying investigation candidates in the k^{th} deliberation.

Given M^k , assume the set of expandable leaves is indexed so that $s_\ell \in L_{\text{exp}}^k$ is the ℓ^{th} leaf in that set. Every expandable leaf is a candidate for investigation. The value of investigating a frontier state s_ℓ is its potential to improve the action choice in the present state.

Since deliberation is costly, the agent must prioritize which frontier states, if any, to investigate. Because the agent unaware of any part of the tree beyond the frontier, she cannot assign ordinary probabilities to what an investigation of s_ℓ might reveal.

Instead, for each expandable frontier state, the agent asks how the continuation values of the feasible actions at her present state s_t could change if the estimated assessment at s_ℓ were refined before she made an action decision. This refinement value is computed by reassessing the present awareness tree twice for each expandable leaf — once at its minimum value and once at its maximum using the following procedure.

If

$$L_{\text{exp}}^k = \emptyset,$$

halt and set

$$k^* := k.$$

Otherwise, recall that leaves are associated with payoff ranges $[\underline{d}(s_\ell), \bar{d}(s_\ell)]$ and, for each $s_\ell \in L_{\text{exp}}^k$, proceed as follows:

1. Begin by assigning s_ℓ its lower endpoint value $v^k(s_\ell) := \underline{d}(s_\ell)$, while leaving all other leaf assignments unchanged.
2. Working in reverse topological order along the path from s_ℓ to s_t , recompute the estimated continuation value at an intermediate state s of its

feasible action $a_i \in A_i(s)$ as

$$v_s^k(a_i) := \sum_{a_N \in A_N(s, a_i)} v^k\left(f_{\text{Depth}(s)}(s, (a_i, a_N))\right) \hat{\sigma}_N(a_N \mid s, a_i).$$

The agent assumes she will choose a subjectively best action upon arrival at any discovered non-leaf state s . Therefore, assign s a continuation value consistent with subjective optimization:

$$v^k(s) := \max_{a_i \in A_i(s)} v_s^k(a_i).$$

Eventually, s_t itself is reached. When that happens, set

$$v_{s_\ell, -}^k(s_t) := v^k(s_t).$$

This denotes the estimated continuation value of s_t obtained by reassessing the discovered awareness tree M^k after temporarily assigning s_ℓ its lower endpoint value $\underline{d}(s_\ell)$. This reassessment is hypothetical; it is used only to decide whether it is worth conducting an investigation at s_ℓ .

3. Next, repeat the same hypothetical reassessment after temporarily assigning s_ℓ its upper endpoint value $\bar{d}(s_\ell)$. This yields

$$v_{s_\ell, +}^k(s_t).$$

4. Restore the temporarily modified assignments $v^k(s)$, $v_s^k(a_i)$, and $A_i^{*,k}(s)$ to their pre-reassessment values.
5. Finally, compute the *marginal benefit of investigating s_ℓ* :

$$B^k(s_\ell) := (1 - \alpha)v_{s_\ell, -}^k(s_t) + \alpha v_{s_\ell, +}^k(s_t) - v^k(s_t).$$

The quantity $B^k(s_\ell)$ measures a subjective value, for the present state- s_t action problem, of being able to refine s_ℓ before acting.

Thus the agent assesses the effects of a frontier state resolving to each of its extremes and then judges whether those values imply an improvement over her present choice of action. At the end of this procedure, all expandable frontier states are assigned a $B^k(\cdot)$ value.

C. Investigation

Upon completing the previous step:

1. Select an expandable leaf with maximal marginal benefit:

$$s_\ell^{k,\star} \in \arg \max_{s_\ell \in L_{\text{exp}}^k} B^k(s_\ell).$$

If there are multiple maximizers, break ties uniformly.

2. Continue deliberation only if the selected leaf's marginal benefit strictly exceeds the marginal cost of the next deliberative round:

$$B^k(s_\ell^{k,\star}) > \Delta\kappa(k+1).$$

If this inequality fails, halt and set

$$k^* := k.$$

Ties are always resolved in favor of stopping.

3. If the inequality holds, undertake a further deliberative round. *Investigate* $s_\ell^{k,\star}$ by revealing its direct successor states, adding them to the awareness tree, and setting

$$\begin{aligned} M^{k+1} &:= M^k \cup \text{Chld}(s_\ell^{k,\star}), \\ L^{k+1} &:= \{s \in M^{k+1} \mid \text{Chld}(s) \cap M^{k+1} = \emptyset\}, \\ L_{\text{exp}}^{k+1} &:= \{s \in L^{k+1} \mid s \notin S_T\}, \end{aligned}$$

and then compute the updated assignments on the enlarged awareness tree M^{k+1} : assign

$$v^{k+1}(s) := \hat{d}(s)$$

to each leaf $s \in L^{k+1}$, compute $v_s^{k+1}(a_i)$ for each feasible action at each discovered non-leaf state, set

$$v^{k+1}(s) := \max_{a_i \in A_i(s)} v_s^{k+1}(a_i),$$

and define the corresponding maximizing-action set $A_i^{*,k+1}(s)$.

4. Set $k := k+1$ and return to the candidate identification routine in Part B.

D. Cleanup

When the deliberation procedure halts, set

$$k^* := k.$$

The final discovered awareness tree and final leaf set are

$$\hat{M}_{s_t}^{\text{fin}} := M^{k^*}, \quad \hat{L}_{s_t}^{\text{fin}} := L^{k^*}.$$

The final assessment at loop step k^* is already available from the last pass through Part C.

For every state $s \in \hat{M}_{s_t}^{\text{fin}}$, define the final estimated continuation value

$$v^*(s) := v^{k^*}(s).$$

Thus final values are available both at discovered non-leaf states and at final frontier states. For every discovered non-leaf state $s \in \hat{M}_{s_t}^{\text{fin}} \setminus \hat{L}_{s_t}^{\text{fin}}$, the deliberation procedure has also computed the final estimated action values

$$v_s^*(a_i) := v_s^{k^*}(a_i), \quad a_i \in A_i(s),$$

and the corresponding set of subjectively optimal actions,

$$A_i^*(s) := \arg \max_{a_i \in A_i(s)} v_s^*(a_i).$$

Action choice

Once deliberation stops, Agent $C+$ chooses an element of the final maximizing set at her present state:

$$a_{C+}(s_t) \in A_i^*(s_t),$$

with ties resolved according to the maintained tie-breaking convention.

Summary. Agent $C+$ differs from Agent C by having the capacity to investigate beyond her initial awareness horizon before acting. But this capacity is disciplined by the present action problem. At each deliberative loop step, the agent asks which presently feasible action looks best at s_t , given the awareness tree currently available to her. She then asks whether investigating any unresolved frontier state has enough expected value for that action problem to justify the marginal cost of another investigation. A frontier state is worth investigating only if refining it can improve the estimated continuation values that matter for the present choice. If no such investigation has positive net value, deliberation stops.

The result is a model of costly awareness expansion rather than unlimited foresight. Agent $C+$ can deliberately expand the awareness tree, update estimated continuation values, and change her intended action when the new information warrants it. But she does not investigate merely because uncertainty remains. Investigation is rational only when it is valuable for choosing among the feasible actions at the state where she must now act.

A.5.3 Agent C^{Plan} : Planning as cached deliberation

Agent C^{Plan} inherits Agent C ’s deliberation procedure (Section A.5.2) but has one additional faculty: she can convert the result of a deliberative episode into a state-contingent plan and then carry that plan forward into later periods. The crucial point is that a plan is not a second deliberation technology, and it is not a stored copy of everything the agent discovered. It is a compressed memory object. It preserves the continuation the agent has resolved to follow, together with the gross estimated continuation-value labels computed for that continuation, but it does not preserve the full deliberative record that led the agent to reject alternative actions.

This distinction is important because planning is not the same as possessing a detailed, long-term memory with which one can recall all the details of a previous deliberation. When the agent first deliberates at a state, she may explore both the action she eventually plans to take and several actions she eventually rejects. Once she instantiates a plan, however, only the on-plan continuation is carried forward. The off-plan branches may have played an essential role in justifying the plan, but they are not themselves part of the plan. Later, when the agent reaches a planned state, those off-plan alternatives may again become locally visible through ordinary arrival awareness. At that point, the agent typically rediscovers, or remembers, aspects of the world she previously considered as part of an earlier deliberation but which did not count as relevant contingencies with respect to her chosen plan. Landing in a state while following a plan and being faced with the reality of her arrival information may cause the agent to reevaluate that plan.

This gives planning a distinctive role in the model. An active plan supplies a zero-deliberative-round input to the present action comparison: the planned action comes with a stored gross estimated continuation value. But there is nothing in the plan that makes reconsideration impossible. Locally visible alternatives may be compared against the planned action through ordinary arrival awareness, and further deliberative rounds may be undertaken when their expected value justifies their marginal cognitive cost. Thus the interesting contrast is not between blind plan-following versus full reoptimization from scratch. It is between acting from a compressed on-plan cache and undertaking additional investigation-analysis rounds to review that plan.

In this sense, Agent C^{Plan} represents a form of rational commitment under costly cognition. Commitment, in the present model, is not a refusal to reconsider, a brute constraint on future choice, or a change in the agent’s underlying preferences. It is the availability of an active plan as a zero-deliberative-round source of action-relevant continuation value, together with the requirement that further investigation and analysis be justified by the expected value of the additional cognition they require.

This refines familiar contrasts between planning and reoptimization. A plan is not treated as a full plan that permanently settles every future contingency, but neither is the agent modeled as if each later decision must be solved again from scratch. Instead, the plan preserves an on-plan continuation and its gross estimated continuation value while leaving rejected alternatives to be reconsidered, extended, or recovered when doing so is worth the present cognitive cost. The model therefore explains why a rational agent may stick with a plan even when an alternative is again visible, why reconsideration need not itself be plan abandonment, and why departure from a plan can be rational rather than a failure of commitment. The formal elaboration below is designed to capture these possibilities while keeping the deliberation machinery the same as that already defined for Agent $C+$.

Features of a plan. The object stored by Agent C^{Plan} is intentionally thinner than the discovered tree from which it is formed. It is not a record of every state the agent has examined, and it is not a full plan over every feasible action at every represented state. It is an on-plan cache: it records the continuation the agent has resolved to follow, the actions prescribed along that continuation, and the gross continuation-value labels computed for the planned states. This is why planning can matter. The agent carries forward the value of the planned continuation without carrying forward the full analysis of all rejected alternatives.

Recall, barred plan notation is used for the stored components of a plan. With that in mind, define a *plan formulated in state* s_t :

$$\bar{\text{PL}}_{s_t} \equiv (\bar{D}_{s_t}, \bar{\sigma}_{s_t}, \bar{\nu}_{s_t}).$$

Here $\bar{D}_{s_t} \subseteq S$ is the plan-domain state set, $\bar{\sigma}_{s_t}$ is the planned state-contingent action prescription, and $\bar{\nu}_{s_t}$ is the stored gross continuation-value function. Thus $\bar{D}_{s_t}$ is a set of within-plan states, while $\bar{\nu}_{s_t}$ is a subjective expected continuation value function defined for such states.¹³

When the agent later reaches a nonterminal frontier state of the incoming plan, the incoming plan is no longer active at that state. The agent then deliberates from ordinary arrival awareness, using the same $C+$ procedure, and forms a new outgoing plan from the resulting final assessment.

The plan-domain set $\bar{D}_{s_t}$ is the set of states associated with a partial subtree of Γ rooted at s_t . It contains the states that could be reached if the agent follows the prescribed action at each within-plan state. The information lost when a plan is

¹³Note: $\bar{\nu}_{s_t}$ is distinct from, though sometimes equivalent to, the scalar proxy value $\hat{d}(s)$ used for unexpanded leaves.

formalized is the full off-plan action-value analysis that originally supported rejection of alternative actions. Therefore, it does not include off-plan states that can only be reached if the agent chooses an action different from the one prescribed by the plan.

The three components of $\bar{\text{PL}}_{s_t}$ satisfy the following conditions:

1. Single planned agent move per planned, non-frontier state: if $s \in \bar{D}_{s_t}$ has at least one successor in $\bar{D}_{s_t}$, then $\bar{\sigma}_{s_t}(s) \in A_i(s)$. The plan therefore tells the agent what to do at planned decision states; it does not store a complete ranking of every feasible action at those states.
2. Nature-closure within-plan: for every non-frontier state $s \in \bar{D}_{s_t}$ and every $a_N \in A_N(s, \bar{\sigma}_{s_t}(s))$,

$$f_{\text{Depth}(s)}(s, (\bar{\sigma}_{s_t}(s), a_N)) \in \bar{D}_{s_t}.$$

This condition ensures that if the agent follows $\bar{\sigma}_{s_t}$ at every planned state, then the realized path remains inside $\bar{D}_{s_t}$ until a plan leaf is reached. The plan is therefore closed under Nature’s feasible responses to the planned action, but it is not closed under the agent’s off-plan actions.¹⁴

3. Stored estimated continuation values: for every on-plan state $s \in \bar{D}_{s_t}$, the plan stores that value as $\bar{v}_{s_t}(s)$.¹⁵ Note: these values are gross — they do not subtract the cognitive cost already incurred in forming the plan. Those past deliberative-round costs are sunk once the plan is formed.

Forming an outgoing plan when no incoming plan is active. At a reached nonterminal decision state s_t , Agent C^{Plan} may have no active incoming plan. There are two ordinary situations in which this is true: initially at s_0 , before any plan has been formed, and later when the agent arrives at a frontier state of the incoming plan.

In such cases, the agent does not use a separate planning technology. She first deliberates exactly as Agent $C+$ deliberates: she begins from arrival awareness and runs the deliberation procedure of Section A.5.2. Suppose this deliberative episode

¹⁴This is a simplifying assumption about what the plan covers. It rules out surprise from an unanticipated Nature move along the planned action. Relaxing this requirement would make the model more realistic, but it is not needed for our present purposes. The point here is narrower: even with Nature-closure, the agent may later reconsider because the plan does not preserve the off-plan deliberative record.

¹⁵The estimated continuation values are computed during the deliberative episode in which the plan is formed.

stops after k^* completed deliberative rounds. The resulting final awareness tree and final leaf set are

$$\hat{M}_{s_t}^{\text{fin}}, \quad \hat{L}_{s_t}^{\text{fin}}.$$

The final assessment has already computed, for each discovered non-leaf state $s \in \hat{M}_{s_t}^{\text{fin}} \setminus \hat{L}_{s_t}^{\text{fin}}$, the estimated continuation values of feasible actions at s

$$v_s^*(a_i), \quad a_i \in A_i(s),$$

the estimated continuation values of the states themselves

$$v^*(s) = \max_{a_i \in A_i(s)} v_s^*(a_i),$$

and the corresponding set of subjectively optimal actions,

$$A_i^*(s) = \arg \max_{a_i \in A_i(s)} v_s^*(a_i).$$

No additional deliberative routine is introduced here; plan formation uses the same final awareness tree and the same estimated continuation values just computed by the $C+$ deliberation procedure.

Then an outgoing plan at s_t is *formulated* by retaining only the on-plan continuation data selected by this final assessment. The within-plan state set $\bar{D}_{s_t}$ is generated by beginning at s_t , assigning a subjectively optimal planned action there, closing under Nature's feasible responses to that planned action, and then repeating this procedure down the awareness tree until the construction reaches final frontier states in $\hat{L}_{s_t}^{\text{fin}}$. Whenever the construction reaches a state $s \in \bar{D}_{s_t} \setminus \hat{L}_{s_t}^{\text{fin}}$, assign as the planned action at s the tie-broken element of the final optimizing set, $\bar{\sigma}_{s_t}(s) \in A_i^*(s)$. Then, add every Nature-successor state reached through that planned action:

$$\bar{D}_{s_t} := \bar{D}_{s_t} \cup \left\{ f_{\text{Depth}(s)}(s, (\bar{\sigma}_{s_t}(s), a_N)) \mid a_N \in A_N(s, \bar{\sigma}_{s_t}(s)) \right\}.$$

The construction then proceeds in the same way from the newly added states. If

$$s \in \bar{D}_{s_t} \cap \hat{L}_{s_t}^{\text{fin}},$$

then s is a frontier state of the plan: no planned action is stored at s , and no successor states beyond s are included in the plan. When the construction is complete, $\bar{D}_{s_t} \subseteq \hat{M}_{s_t}^{\text{fin}}$.

Finally, for each within-plan state $s \in \bar{D}_{s_t}$, assign the plan continuation value computed during the final assessment:

$$\bar{v}_{s_t}(s) := v^*(s).$$

Again, these are gross values — they do not subtract the (sunk) cognitive cost already incurred in the deliberative episode that produced the plan. This yields the plan

$$\bar{\text{PL}}_{s_t} \equiv (\bar{D}_{s_t}, \bar{\sigma}_{s_t}, \bar{\nu}_{s_t}).$$

This plan is typically much more sparse than the awareness tree from which it was formed. Even if the agent investigated off-plan branches while deciding what to do at s_t , those rejected branches are not carried forward as part of $\bar{\text{PL}}_{s_t}$. What *is* carried forward is the planned continuation, the action prescription along that continuation, and the stored gross estimated continuation values on that continuation. With this plan in hand, the agent implements the present prescribed action $\bar{\sigma}_{s_t}(s_t)$. Nature then selects its move, and the process transitions to the next realized state.

Arrival with an incoming plan. The preceding discussion covered the case in which Agent C^{Plan} arrives in s_t with no active incoming plan. The more typical case arises later, after a plan has already been formed and the agent arrives at s_t as a within-plan state as part of an incoming plan

$$\bar{\text{PL}}_{s_{t'}} = (\bar{D}_{s_{t'}}, \bar{\sigma}_{s_{t'}}, \bar{\nu}_{s_{t'}}),$$

that was formulated at the earlier state $s_{t'}$, $t' < t$. In such cases, the first question is whether that prior plan is still usable at the state the agent has actually reached. The incoming plan $\bar{\text{PL}}_{s_{t'}}$ *is active at s_t* if and only if

$$s_t \in \bar{D}_{s_{t'}}, \quad \bar{\sigma}_{s_{t'}}(s_t) \text{ is defined}, \quad \bar{\sigma}_{s_{t'}}(s_t) \in A_i(s_t), \quad \bar{\nu}_{s_{t'}}(s_t) \text{ is defined}.$$

This definition is deliberately modest. It does not say that an active plan contains every action available to the agent at s_t . Nor does it say that the prior plan domain may be treated as a full discovered tree for fresh backward induction. It says only that the plan supplies a usable state-contingent action map and stored continuation values for within-plan states. When the plan is active, the agent can follow it without incurring any positive deliberative-round cost by choosing $\bar{\sigma}_{s_{t'}}(s_t)$ and relying on $\bar{\nu}_{s_{t'}}(s_t)$ as the previously computed continuation value of that planned choice.

It is useful to name the portion of the old plan that remains relevant from the reached state onward. When a plan formulated at $s_{t'}$ is active at s_t , define *the plan continuation at s_t* as

$$\bar{P}_{s_t} := \bar{D}_{s_{t'}} \cap (\text{Desc}(s_t) \cup \{s_t\}).$$

The set $\bar{P}_{s_t}$ is the set of within-plan continuation states from s_t — the states of the partial subtree of Γ rooted at s_t that are retained by the active plan. It is not

the agent's whole present awareness tree upon landing at s_t because ordinary arrival awareness *will* make locally available off-plan children visible again.¹⁶ Those off-plan children correspond to alternatives that were examined and rejected when the plan was first formed and, hence, not retained as part of the plan itself.¹⁷

Deliberation with an active incoming plan. Suppose a plan formulated at $s_{t'}$ is active upon arrival at s_t , $t' < t$. The agent receives ordinary arrival awareness there. In addition, the active plan supplies the planned action $\bar{\sigma}_{s_{t'}}(s_t)$ and the stored estimated continuation value $\bar{v}_{s_{t'}}(s_t)$ for that planned action. The aim of deliberation at s_t is still the same: the agent must now choose among the feasible actions at s_t . What differs under an active plan is the informational basis on which those action values are now estimated.

Arrival initialization proceeds just as $C+$ with the exception that we adjust it to account for the presence of an active plan. Create the *off-plan action set at arrival state s_t* as

$$A_i^{-P}(s_t) := A_i(s_t) \setminus \{a_t^P\},$$

where $a_t^P \equiv \bar{\sigma}_{s_{t'}}(s_t)$ denotes the action prescribed by the active plan at s_t (to avoid notational clutter). Then create the off-plan arrival awareness set:

$$\hat{M}_{s_t}^{\text{arr}, -P} := \{s_t\} \cup \{f_{\text{Depth}(s_t)}(s_t, (a_i, a_N)) \mid a_i \in A_i^{-P}(s_t), a_N \in A_N(s_t, a_i)\}.$$

Now initialize

$$\begin{aligned} M^0 &:= \hat{M}_{s_t}^{\text{arr}, -P}, \\ L^0 &:= \{s \in M^0 \mid \text{Chld}(s) \cap M^0 = \emptyset\}, \text{ and} \\ L_{\text{exp}}^0 &:= \{s \in L^0 \mid s \notin S_T\}. \end{aligned}$$

For each off-plan leaf $s \in L^0$, set the continuation value exactly as before:

$$v^0(s) := \hat{d}(s) = (1 - \alpha)\underline{d}(s) + \alpha\bar{d}(s).$$

For each feasible off-plan action $a_i \in A_i(s_t)$, $a_i \neq a_t^P$, compute

$$v_{s_t}^0(a_i) := \sum_{a_N \in A_N(s_t, a_i)} v^0\left(f_{\text{Depth}(s_t)}(s_t, (a_i, a_N))\right) \hat{\sigma}_N(a_N \mid s_t, a_i).$$

¹⁶Recall, every state has at least two feasible actions for the agent, only one of which is assigned to the plan.

¹⁷Being a within-plan state implies that s_t was investigated earlier during a state $s_{t'}$ deliberation sequence. As such, all the children of s_t were revealed. During plan formulation, only the continuation states on paths through the planned action $\bar{\sigma}_{s_{t'}}(s_t)$ were retained.

For the planned action, set

$$v_{s_t}^0(a_t^P) := \bar{v}_{s_{t'}}(s_t).$$

Finally, the initial estimated continuation value of s_t is

$$v^0(s_t) = \max \left\{ \bar{v}_{s_{t'}}(s_t), \max_{a_i \in A_i^{-P}(s_t)} v_{s_t}^0(a_i) \right\}.$$

From here, the agent conducts deliberation on off-plan paths using the same loop as Agent $C+$. At each review step k , the expandable frontier states are the expandable leaves of the current off-plan review tree M^k . For those leaves, the agent computes $B^k(s_\ell)$, compares the best available $B^k(s_\ell)$ with the marginal cost of the next deliberative round, and either stops or investigates the selected off-plan frontier state. Throughout this review, the planned action a_t^P keeps its cached value $\bar{v}_{s_{t'}}(s_t)$; positive review refines only off-plan alternatives to the active incoming plan.

Throughout this review, the planned action a_t^P keeps its cached value $\bar{v}_{s_{t'}}(s_t)$; positive review refines only off-plan alternatives to the active incoming plan. If the final assessment selects an off-plan feasible action at s_t , then a_t^P is displaced and the agent formulates a new outgoing plan in the same fashion as she did when she arrived with no active incoming plan.¹⁸ If the planned action is not displaced, then the agent follows the active incoming plan at s_t , and the relevant continuation of that plan remains the outgoing plan from s_t .

Commitment interpretation. The preceding procedure gives the local structure of rational commitment under costly cognition. An active plan can persist because it places a cached gross estimated continuation value for the planned action into the present action comparison at zero deliberative rounds. But the plan does not give the planned action an independent payoff bonus, and it does not make the planned action immune from comparison.

A locally visible off-plan action may defeat the planned action at zero rounds if the zero-round action-value assessment selects it. Likewise, positive deliberative rounds may confirm or replace the active plan whenever the inherited $C+$ stopping rule makes further investigation worthwhile. The distinctive commitment case arises when the planned action remains selected under the present assessment and the best available further investigation does not justify its marginal deliberative-round cost.

¹⁸Following review, ties may arise between the present planned action and some alternative feasible actions. In such cases, the usual tie-breaking convention is dropped: the planned action *always* wins when tied with other actions following a review.

Then the agent follows the active plan, not because she refuses to reconsider, but because ordinary action-value maximization under costly deliberation gives her no reason to spend the next deliberative round.

Thus commitment and flexibility are not opposites in the model. A plan guides action by preserving deliberative work in usable form: it carries forward an on-plan continuation and its gross estimated continuation values. At the same time, the agent remains responsive to locally visible alternatives and to positive-value opportunities for further investigation. Agent C^{Plan} therefore occupies the intermediate case: she need not solve the same continuation problem from scratch at every later state, but she may recheck, depart from, or replace the plan when the inherited deliberation procedure makes doing so worthwhile. Planning matters not because the model assigns an independent preference for prior plans, but because an active plan changes the agent's informational and computational position at later states.

Confirmatory depth and plan persistence. Deeper prior exploration does not unconditionally make a plan more persistent. It may lower the plan's continuation assessment, change its prescription, or expose considerations that make renewed deliberation more valuable. A monotonic result becomes available when deeper exploration is *confirmatory*: it resolves additional uncertainty while preserving the plan's prescription and its current scalar assessment.

To make this precise, compare two incoming plans at the same reached state s_t . Call them the *shallow plan* and the *deep plan*. Suppose the shallow plan has an unexpanded retained frontier state s_f with stored interval

$$\bar{I}^{\text{sh}}(s_f) = [L_f, U_f]$$

and scalar assessment

$$m_f := (1 - \alpha)L_f + \alpha U_f.$$

During formation of the deep plan, the agent has explored beyond s_f , selected an on-plan continuation there, and retained the resulting subtree. Let the interval obtained by applying the plan-interval recursion to that deeper continuation be

$$\bar{I}^{\text{dp}}(s_f) = [L'_f, U'_f].$$

Call this a *confirmatory expansion at s_f* if

$$[L'_f, U'_f] \subseteq [L_f, U_f]$$

and

$$(1 - \alpha)L'_f + \alpha U'_f = m_f.$$

Thus the deeper exploration narrows the unresolved range without changing the α -assessment propagated through the plan. Assume throughout that $0 < \alpha < 1$, that the two plans have the same prescriptions and credences at all common upstream states, and that their ordinary $C+$ representations, current action set, tie-breaking convention, and deliberation-cost schedule are the same.

For each plan, define the best initial refinement value by

$$\beta^j := \max \left(\{B_j^0(D, s_\ell) \mid (D, s_\ell) \in \mathcal{E}_j^0\} \cup \{0\} \right), \quad j \in \{\text{sh}, \text{dp}\}.$$

The added zero covers the case in which the relevant candidate set is empty.

Proposition A.5.1 (Confirmatory depth weakly increases plan persistence). *Suppose the deep incoming plan is obtained from the shallow incoming plan by a confirmatory expansion at s_f , with every other retained-plan frontier and every ordinary $C+$ representation unchanged. Then:*

1. *The two plan-enabled episodes have the same initial deliberation choice values, physical action values, and maximizing physical actions.*
2. *Every initial refinement candidate unaffected by the expansion has the same refinement value under the two plans. Every new retained-plan frontier candidate below s_f in the deep plan has refinement value no greater than the refinement value of s_f in the shallow plan.*
3. *Consequently,*

$$\beta^{\text{dp}} \leq \beta^{\text{sh}}.$$

If, in addition, the planned action is uniquely maximizing and uniquely supported by the retained-plan representation at $k = 0$, then immediate retention of the shallow plan implies immediate retention of the deep plan. More generally, if

$$\beta^{\text{dp}} \leq \Delta\kappa(1) < \beta^{\text{sh}},$$

the shallow plan induces renewed deliberation while the deep plan is retained without further investigation.

Proof. At every newly retained frontier below s_f , the scalar proxy is the α -assessment of its stored interval. The plan-interval recursion is affine in those frontier values because the retained continuation fixes the agent's prescriptions and aggregates Nature's responses using fixed credences. The confirmatory condition therefore implies that the value propagated from s_f through every common upstream plan state is the

same under the shallow and deep plans. All ordinary representations are unchanged. Hence every initial deliberation-choice value and every physical action value is the same under the two plans. This proves Part 1.

Hold all other initial values fixed, and let $\Phi_f(x)$ be the value assigned to the current state s_t when the value propagated from s_f through the retained-plan representation is x . Upstream propagation within that representation is affine and nondecreasing in x . At the current state, the resulting plan-supported action value is compared with the values supplied by the other deliberation choices. It follows that Φ_f is the maximum of finitely many affine functions of x , and is therefore convex and nondecreasing.

For the shallow plan, the refinement value of s_f can be written

$$B_{\text{sh}}^0(D^P(a_t^P), s_f) = (1 - \alpha)\Phi_f(L_f) + \alpha\Phi_f(U_f) - \Phi_f(m_f).$$

Now consider any new retained-plan frontier r below s_f in the deep plan. Let y_r^- and y_r^+ denote the values propagated to s_f when r is assigned, respectively, the lower and upper endpoint of its stored interval, while every other deep-plan frontier retains its scalar proxy. Affinity of the plan recursion and the common parameter α imply

$$(1 - \alpha)y_r^- + \alpha y_r^+ = m_f.$$

Because every other frontier proxy lies between its own endpoints,

$$L_f' \leq y_r^- \leq y_r^+ \leq U_f'.$$

The nesting condition therefore gives

$$L_f \leq y_r^- \leq y_r^+ \leq U_f.$$

The two-point distribution placing probabilities $1 - \alpha$ and α on y_r^- and y_r^+ is consequently a mean-preserving contraction of the corresponding distribution on L_f and U_f . Convexity of Φ_f yields

$$\begin{aligned} B_{\text{dp}}^0(D^P(a_t^P), r) &= (1 - \alpha)\Phi_f(y_r^-) + \alpha\Phi_f(y_r^+) - \Phi_f(m_f) \\ &\leq (1 - \alpha)\Phi_f(L_f) + \alpha\Phi_f(U_f) - \Phi_f(m_f) \\ &= B_{\text{sh}}^0(D^P(a_t^P), s_f). \end{aligned}$$

Candidates outside the newly retained subtree have unchanged endpoints, baseline values, and induced action comparisons, so their refinement values are unchanged. The shallow candidate at s_f is removed and replaced only by descendant

candidates whose refinement values are no larger. Taking the maximum proves Parts 2 and 3.

Finally, Part 1 preserves the unique initial support for the planned action. If the shallow plan satisfies $\beta^{\text{sh}} \leq \Delta\kappa(1)$, then Part 3 implies $\beta^{\text{dp}} \leq \Delta\kappa(1)$, so both procedures stop immediately and retain their incoming plans. If the displayed strict threshold condition holds, the stopping rule instead makes the shallow procedure investigate and the deep procedure stop at $k = 0$. $\square$

Corollary A.5.2 (Persistence under iterated confirmatory exploration). *Consider a finite sequence of incoming plans in which each plan is obtained from its predecessor by one or more confirmatory expansions satisfying the conditions of Proposition A.5.1. Let $\beta^0, \beta^1, \dots, \beta^n$ denote their best initial refinement values. Then*

$$\beta^0 \geq \beta^1 \geq \dots \geq \beta^n.$$

If the planned action remains uniquely maximizing and uniquely supported by the retained-plan representation throughout the sequence, then once some plan in the sequence is retained without further investigation, every subsequent, more deeply confirmed plan is also retained without further investigation.

Proof. Apply Proposition A.5.1 successively to each adjacent pair of plans. The monotonicity of the stopping result follows from the common first-round cost threshold. $\square$

When the inequality is strict. The root-value function Φ_f is typically piecewise linear. Strict reduction in refinement value occurs when the shallower interval crosses a kink at which the maximizing physical action or supporting deliberation choice changes, while the confirmatory expansion reduces the possibility of crossing that kink. If Φ_f is affine throughout the relevant interval, endpoint refinement has no option value there, and narrowing need not change the refinement calculation. The economically important strict case is therefore one in which the shallow plan leaves open a genuine possibility of reversal and deeper exploration removes enough of that possibility to cross the deliberation-cost threshold.

Interpretation and limits. The proposition identifies a precise sense in which carefully formed plans are more persistent. Deeper confirmatory exploration does not increase the stored plan’s scalar value and does not give its action motivational priority. It reduces the value of reopening questions that the earlier deliberation has already partly resolved. The plan becomes more cognitively persistent because fewer unresolved contingencies can profitably alter the current choice.

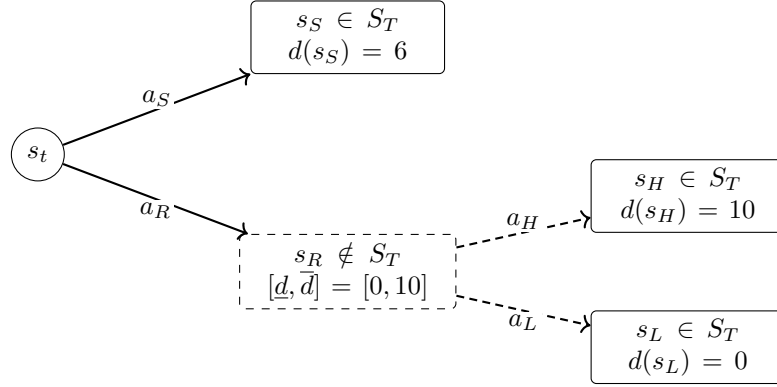


Figure A.4: A one-round Agent $C+$ deliberation problem. At $k = 0$, the agent is aware of s_t , s_S , and s_R . The dashed continuations from s_R are revealed only if s_R is investigated.

This is not a theorem that depth always produces persistence. The nesting and scalar-preservation conditions do substantive work. If deeper exploration moves the plan interval downward, changes the prescribed continuation, alters the relevant credences, or exposes a sufficiently important new comparison, the initial action values and refinement opportunities can change. Deeper exploration may then destabilize or displace the plan. That possibility is consistent with the model's interpretation: further inquiry should strengthen a plan when it confirms the earlier assessment and weaken the plan when it uncovers reasons for revision.

The result therefore avoids treating persistence as brute stickiness. The greater stability of a deeply confirmed plan is earned by the uncertainty resolved during its formation. Its authority is neither unconditional nor purely motivational; it is the continuing practical effect of earlier deliberative work.

A.6 Examples

A.6.1 Agent $C+$: one-round value of refinement

Example 3 (One-round value of refinement). *Consider the decision problem in Figure A.4. At the current state s_t , Agent $C+$ has two feasible actions, $A_i(s_t) = \{a_S, a_R\}$. The action a_S is safe: after the singleton Nature response a_N^S , it leads to a terminal state s_S with $d(s_S) = 6$:*

$$f_{\text{Depth}(s_t)}(s_t, (a_S, a_N^S)) = s_S.$$

The action a_R is risky: after the singleton Nature response a_N^R , it leads to a nonterminal state s_R :

$$f_{\text{Depth}(s_t)}(s_t, (a_R, a_N^R)) = s_R.$$

All singleton Nature responses in the example are assigned probability one.

If the agent investigates s_R , she discovers two feasible future actions, a_H and a_L , again with singleton Nature responses:

$$f_{\text{Depth}(s_R)}(s_R, (a_H, a_N^H)) = s_H, \quad f_{\text{Depth}(s_R)}(s_R, (a_L, a_N^L)) = s_L,$$

where $s_H, s_L \in S_T$, $d(s_H) = 10$, and $d(s_L) = 0$.

At arrival, however, s_H and s_L are not yet in the agent's awareness tree. The agent knows only that the continuation through s_R lies in the interval $[\underline{d}(s_R), \bar{d}(s_R)] = [0, 10]$. Set $\alpha = \frac{1}{2}$. Thus the point estimate of the unresolved leaf s_R is

$$\hat{d}(s_R) = (1 - \alpha)\underline{d}(s_R) + \alpha\bar{d}(s_R) = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 10 = 5.$$

The initial discovered tree and leaf sets are

$$M^0 = \hat{M}_{s_t}^{\text{arr}} = \{s_t, s_S, s_R\}, \quad L^0 = \{s_S, s_R\}, \quad L_{\text{exp}}^0 = \{s_R\}.$$

Here s_S is terminal, while s_R is a nonterminal expandable leaf.

The baseline $k = 0$ assessment assigns $v^0(s_S) = 6$ and $v^0(s_R) = \hat{d}(s_R) = 5$. Because Nature's responses at s_t are singleton responses, $v_{s_t}^0(a_S) = 6$ and $v_{s_t}^0(a_R) = 5$. Hence $v^0(s_t) = 6$ and $A_i^{*,0}(s_t) = \{a_S\}$. So without further deliberation, Agent C+ would choose the safe action a_S .

Now consider the marginal benefit of investigating the only expandable leaf s_R . For the lower reassessment, temporarily assign $v^0(s_R) := \underline{d}(s_R) = 0$. Then the current action values are $v_{s_t}^0(a_S) = 6$ and $v_{s_t}^0(a_R) = 0$, so the reassessed value of the present state is $v_{s_R,-}^0(s_t) = 6$. For the upper reassessment, temporarily assign $v^0(s_R) := \bar{d}(s_R) = 10$. Then the current action values are $v_{s_t}^0(a_S) = 6$ and $v_{s_t}^0(a_R) = 10$, so $v_{s_R,+}^0(s_t) = 10$. Restoring the baseline assignments, the value of refinement is

$$B^0(s_R) = (1 - \alpha)v_{s_R,-}^0(s_t) + \alpha v_{s_R,+}^0(s_t) - v^0(s_t).$$

With $\alpha = \frac{1}{2}$,

$$B^0(s_R) = \frac{1}{2} \cdot 6 + \frac{1}{2} \cdot 10 - 6 = 2.$$

Assume the first marginal deliberation cost is $\Delta\kappa(1) = 1$. Since $B^0(s_R) = 2 > \Delta\kappa(1) = 1$, Agent C+ undertakes one deliberative round and investigates s_R .

Investigation reveals the direct successors of s_R , so

$$M^1 = \{s_t, s_S, s_R, s_H, s_L\}, \quad L^1 = \{s_S, s_H, s_L\}, \quad L_{\text{exp}}^1 = \emptyset.$$

The terminal values are $v^1(s_S) = 6$, $v^1(s_H) = 10$, and $v^1(s_L) = 0$. At the newly discovered non-leaf state s_R , the action values are $v_{s_R}^1(a_H) = 10$ and $v_{s_R}^1(a_L) = 0$. Thus $v^1(s_R) = 10$ and $A_i^{*,1}(s_R) = \{a_H\}$.

Working back to the current state s_t , the updated action values are $v_{s_t}^1(a_S) = 6$ and $v_{s_t}^1(a_R) = 10$. Therefore $v^1(s_t) = 10$ and $A_i^{*,1}(s_t) = \{a_R\}$.

Since $L_{\text{exp}}^1 = \emptyset$, the deliberation procedure halts with $k^* = 1$. The final discovered awareness tree and final leaf set are

$$\hat{M}_{s_t}^{\text{fin}} = M^1, \quad \hat{L}_{s_t}^{\text{fin}} = L^1.$$

The final maximizing-action set at s_t is $A_i^*(s_t) = \{a_R\}$. Hence Agent $C+$ chooses $a_{C+}(s_t) = a_R$, with gross final continuation value $v^*(s_t) = 10$.

The point of the example is that the agent does not investigate because she has an exogenous search capacity. She investigates because the value of refining s_R for the present action problem strictly exceeds the marginal cost of the next deliberative round.

A.6.2 Agent C^{Plan} : cached values and off-plan review

The next two examples use the same local decision structure—an active incoming plan at s_t facing a single locally visible off-plan alternative—but differ in the quality of that alternative and in the marginal cost of reviewing it. The point is to separate two roles of planning. In the first example the off-plan branch is genuinely inferior to the cached continuation, and the active plan lets the agent avoid a locally tempting but worse alternative at zero deliberative cost. In the second the off-plan branch is genuinely superior, and the same review procedure rationally displaces the plan once off-plan investigation is worth its cost.

Example 4 (Planning advantage from a cached continuation value). *Consider the local decision problem in Figure A.5. Agent C^{Plan} arrives at s_t with an active incoming plan $\bar{\text{P}}_{s_{t'}}$, where $t' < t$. In the deliberative episode that formed the plan, the agent resolved the planned continuation and cached its value, so the plan prescribes $a_t^P = \bar{\sigma}_{s_{t'}}(s_t)$ and stores the gross continuation value $\bar{v}_{s_{t'}}(s_t) = 10$. There is one locally visible off-plan action a_O , so $A_i^{-P}(s_t) = \{a_O\}$. The off-plan action leads, after a singleton Nature response, to an unresolved nonterminal leaf r with $[\underline{d}(r), \bar{d}(r)] = [6, 8]$. Set $\alpha = \frac{1}{2}$, so $\hat{d}(r) = 7$.*

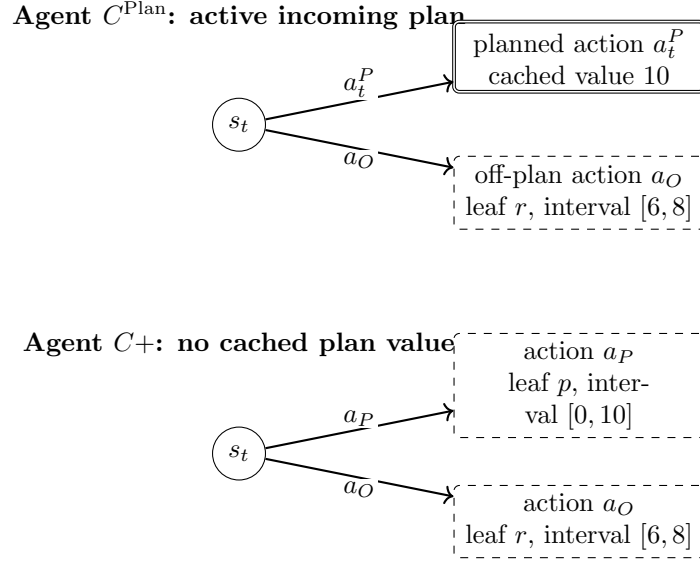


Figure A.5: A local comparison between Agent C^{Plan} , who arrives with an active incoming plan, and Agent $C+$, who lacks the cached value of the planned continuation. The off-plan interval $[6, 8]$ lies entirely below the cached value 10, so the planned action is genuinely the better choice. The top panel shows C^{Plan} 's cached planned action and the locally visible off-plan alternative; the bottom panel shows the corresponding ordinary arrival-awareness problem for $C+$, who sees only the crude interval $[0, 10]$ for the planned branch.

The active-plan review tree begins from the off-plan arrival-awareness set $M^0 = \hat{M}_{s_t}^{\text{arr}, -P} = \{s_t, r\}$. The planned action is not expanded inside this off-plan review tree. Instead, it keeps its cached value:

$$v_{s_t}^0(a_t^P) = \bar{v}_{s_{t'}}(s_t) = 10.$$

The off-plan action has initial value $v_{s_t}^0(a_O) = \hat{d}(r) = 7$. Thus $v^0(s_t) = 10$, and the zero-round maximizing set is $A_i^{*,0}(s_t) = \{a_t^P\}$.

Now compute the value of reviewing the off-plan leaf r . The cached value already exceeds both endpoints of the off-plan interval, so neither reassessment can dislodge the planned action. Under the lower reassessment, $v_{s_t}^0(a_t^P) = 10$ and $v_{s_t}^0(a_O) = 6$, so $v_{r,-}^0(s_t) = 10$. Under the upper reassessment, $v_{s_t}^0(a_t^P) = 10$ and $v_{s_t}^0(a_O) = 8$, so $v_{r,+}^0(s_t) = 10$. Restoring the baseline assignments,

$$B^0(r) = (1 - \alpha)v_{r,-}^0(s_t) + \alpha v_{r,+}^0(s_t) - v^0(s_t) = \frac{1}{2} \cdot 10 + \frac{1}{2} \cdot 10 - 10 = 0.$$

Assume the first marginal review cost is $\Delta\kappa(1) = 3$. Since $B^0(r) = 0$, the review carries no value at any positive cost, so a fortiori $B^0(r) < \Delta\kappa(1)$ and Agent C^{Plan} stops at zero review rounds. Because the zero-round assessment selects the planned action, she follows the active incoming plan and chooses $a_{C^{\text{Plan}}}(s_t) = a_t^P$.

Compare this with Agent $C+$ at the same reached state. Agent $C+$ has no cached value for the planned continuation. Suppose ordinary arrival awareness gives $C+$ the initial tree $M_{C+}^0 = \{s_t, p, r\}$, where action a_P leads to the unresolved leaf p with interval $[0, 10]$ —the same continuation the active plan resolved to the cached value 10—while action a_O leads to the unresolved leaf r with interval $[6, 8]$. With $\alpha = \frac{1}{2}$, the zero-round estimates are $\hat{d}(p) = 5$ and $\hat{d}(r) = 7$. Thus $v_{s_t}^0(a_P) = 5$, $v_{s_t}^0(a_O) = 7$, and $A_i^{*,0}(s_t) = \{a_O\}$.

The value of refining the planned branch p for Agent $C+$ is

$$B^0(p) = \frac{1}{2} \cdot 7 + \frac{1}{2} \cdot 10 - 7 = 1.5,$$

while the value of refining r is

$$B^0(r) = \frac{1}{2} \cdot 6 + \frac{1}{2} \cdot 8 - 7 = 0.$$

With the marginal cost $\Delta\kappa(1) = 3$, the best available refinement $B^0(p) = 1.5$ does not justify a deliberative round, so Agent $C+$ also stops at zero rounds—and chooses a_O rather than a_P .

The comparison is local to the reached decision state s_t , but here the planned action is not merely the one the cache happens to favor; it is in fact the better choice. Because $\bar{d}(r) = 8$ is the largest payoff reachable through the off-plan branch, the off-plan action is worth at most 8, whereas the resolved planned continuation is worth 10. Agent C^{Plan} realizes this larger value because the active incoming plan supplies the resolved continuation value at zero deliberative rounds. Agent $C+$, lacking that cached value, sees only the crude interval $[0, 10]$ for the planned branch, undervalues it at its midpoint 5, and is drawn to the locally higher off-plan estimate 7—settling for an action worth 8 instead of 10. Notably, the refinement that would have rescued $C+$ is investigation of the planned branch p , the only leaf with positive marginal benefit; had her marginal cost fallen below 1.5, she would have expanded p , discovered its value, and recovered the plan. The cache spares Agent C^{Plan} exactly that cost.

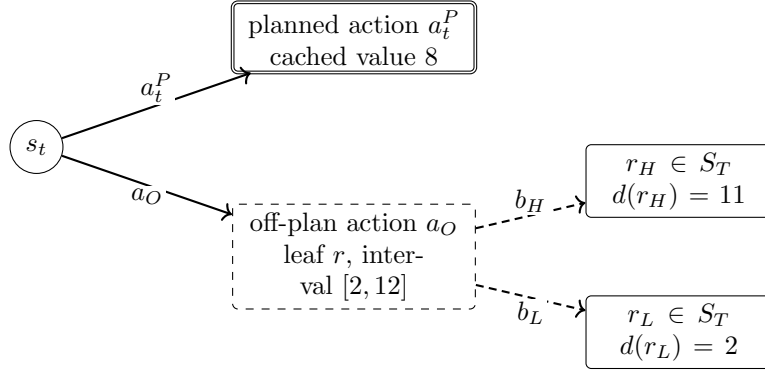


Figure A.6: Plan abandonment after off-plan review. The planned action keeps its cached value during review. The dashed continuations from r are not part of the active incoming plan; they are revealed only if Agent C^{Plan} undertakes a positive review round.

Example 5 (Plan abandonment after off-plan review). *Now keep the same active incoming plan and the same zero-round off-plan estimate, but lower the first marginal review cost. Agent C^{Plan} again arrives at s_t with an active incoming plan satisfying $a_t^P = \bar{\sigma}_{s_t}(s_t)$ and $\bar{v}_{s_t}(s_t) = 8$. The locally visible off-plan action is a_O , and the off-plan review awareness set is again $M^0 = \hat{M}_{s_t}^{\text{arr}, -P} = \{s_t, r\}$, where $r \notin S_T$ has interval $[2, 12]$. With $\alpha = \frac{1}{2}$, the off-plan estimate remains $\hat{d}(r) = 7$.*

At zero review rounds, the planned action still wins the present comparison: $v_{s_t}^0(a_t^P) = 8$, $v_{s_t}^0(a_O) = 7$, $v_{s_t}^0(s_t) = 8$, and $A_t^{,0}(s_t) = \{a_t^P\}$. The value-of-refinement calculation is the same as before:*

$$B^0(r) = \frac{1}{2} \cdot 8 + \frac{1}{2} \cdot 12 - 8 = 2.$$

Now assume $\Delta\kappa(1) = 1$. Since $B^0(r) = 2 > 1 = \Delta\kappa(1)$, Agent C^{Plan} undertakes one positive review round and investigates the off-plan leaf r .

Investigation reveals two terminal successors of r , as shown in Figure A.6. The successor r_H has terminal value $d(r_H) = 11$, while r_L has terminal value $d(r_L) = 2$. Suppose the agent's feasible continuation actions at r are b_H and b_L , with singleton Nature responses leading to r_H and r_L , respectively. After the review round, the enlarged off-plan review tree is $M^1 = \{s_t, r, r_H, r_L\}$, with $L^1 = \{r_H, r_L\}$ and $L_{\text{exp}}^1 = \emptyset$.

The updated terminal values are $v^1(r_H) = 11$ and $v^1(r_L) = 2$. At the newly discovered non-leaf r , the action values are $v_r^1(b_H) = 11$ and $v_r^1(b_L) = 2$, so $v^1(r) = 11$ and $A_t^{,1}(r) = \{b_H\}$. Working back to s_t , the off-plan action now has value $v_{s_t}^1(a_O) = 11$. The planned action continues to use its cached value, so $v_{s_t}^1(a_t^P) = 8$.*

Thus the final review assessment gives $v^1(s_t) = 11$ and $A_i^{*,1}(s_t) = \{a_O\}$. Since $L_{\text{exp}}^1 = \emptyset$, review stops with $k^* = 1$. The off-plan action strictly displaces the planned action, so Agent C^{Plan} abandons the active incoming plan at s_t and chooses $a_{C^{\text{Plan}}}(s_t) = a_O$.

The agent then formulates a new outgoing plan from the reviewed tree in the same way she would when arriving with no active incoming plan. The important point is that plan abandonment is not modeled as a failure of commitment. It occurs because the inherited $C+$ review procedure makes further investigation worthwhile, and the resulting off-plan action value strictly exceeds the cached value of the planned action.

A.7 Tables of notation

The quick index below maps each symbol to the table where it is defined. The tables are organized to match the exposition: Basic Elements, then the notation introduced by each agent family.

Quick lookup (alphabetical): symbol $\rightarrow$ where defined

$A(s)$	Basic: Tbl. A.1 , p. 49	a_t	Basic: Tbl. A.1 , p. 49
$A_i^{-P}(s_t)$	Agent C^{Plan} : Tbl. A.5 , p. 53	$A_i^{*,k}(s)$	Agent C : Tbl. A.4 , p. 51
$A_i^*(s)$	Agent C : Tbl. A.4 , p. 51	a_t^P	Agent C^{Plan} : Tbl. A.5 , p. 53
$a_{C+}(s_t)$	Agent C : Tbl. A.4 , p. 51	$a_{C^{\text{Plan}}}(s_t)$	Agent C^{Plan} : Tbl. A.5 , p. 53
$B^k(s_\ell)$	Agent C : Tbl. A.4 , p. 51	$\bar{D}_{s_t}$	Agent C^{Plan} : Tbl. A.5 , p. 53
$\bar{\sigma}_{s_t}$	Agent C^{Plan} : Tbl. A.5 , p. 53	$\bar{v}_{s_t}$	Agent C^{Plan} : Tbl. A.5 , p. 53
$\bar{P}_{s_t}$	Agent C^{Plan} : Tbl. A.5 , p. 53	$d(\cdot)$	Agent A : Tbl. A.2 , p. 50
$\Delta\kappa(m+1)$	Agent C : Tbl. A.4 , p. 51	$\text{Desc}(s)$	Agent C : Tbl. A.4 , p. 51
$\mathbb{E}(\sigma)$	Agent A : Tbl. A.2 , p. 50	f_t	Basic: Tbl. A.1 , p. 49
Γ	Basic: Tbl. A.1 , p. 49	$\hat{d}(s)$	Agent C : Tbl. A.4 , p. 51
$\hat{L}_{s_t}^{\text{fin}}$	Agent C : Tbl. A.4 , p. 51	$\hat{M}_{s_t}^{\text{arr}}$	Agent C : Tbl. A.4 , p. 51
$\hat{M}_{s_t}^{\text{arr},-P}$	Agent C^{Plan} : Tbl. A.5 , p. 53	$\hat{M}_{s_t}^{\text{fin}}$	Agent C : Tbl. A.4 , p. 51
$\hat{\sigma}_N$	Agent B : Tbl. A.3 , p. 50	$\hat{\sigma}_N(\cdot \mid h_t)$	Agent B : Tbl. A.3 , p. 50
$h(s_t)$	Basic: Tbl. A.1 , p. 49	$\iota(\bar{\sigma}_N, h_t)$	Agent B : Tbl. A.3 , p. 50
k^*	Agent C : Tbl. A.4 , p. 51	$\kappa(m)$	Agent C : Tbl. A.4 , p. 51
L^k, L_{exp}^k	Agent C : Tbl. A.4 , p. 51	M^k	Agent C : Tbl. A.4 , p. 51
$\bar{\text{PL}}_{s_t}$	Agent C^{Plan} : Tbl. A.5 , p. 53	$\text{Par}(s)$	Agent C : Tbl. A.4 , p. 51
p_σ	Basic: Tbl. A.1 , p. 49	$\preceq, \preccurlyeq$	Agent A : Tbl. A.2 , p. 50
s_t, S_t, S	Basic: Tbl. A.1 , p. 49	$s_\ell^{k,*}$	Agent C : Tbl. A.4 , p. 51
$\sigma_i, \sigma_N, \Sigma$	Basic: Tbl. A.1 , p. 49	$\underline{d}(s), \bar{d}(s)$	Agent C : Tbl. A.4 , p. 51
$v^k(s)$	Agent C : Tbl. A.4 , p. 51	$v_s^k(a_i)$	Agent C : Tbl. A.4 , p. 51
$v_{s_\ell,-}^k(s_t), v_{s_\ell,+}^k(s_t)$	Agent C : Tbl. A.4 , p. 51	$v^*(s), v_s^*(a_i)$	Agent C : Tbl. A.4 , p. 51

Table A.1. Basic elements of the theory (used throughout)*States, time, and histories*

s_t	State of the world at time t .
S_t	Set of possible states at time t .
S	Full state space, $S = \bigcup_{t=0}^T S_t$.
s_0	Root state (initial state).
$h(s_t)$	History leading to s_t : $(s_0, \dots, s_t)$.
H_t	Set of period- t histories.
H	Set of all histories, $H = \bigcup_{t=0}^T H_t$.

Actions and transitions

$A_i(s)$	Feasible actions for agent i in state s .
$A_N(s, a_i)$	Feasible Nature actions in s after agent chooses a_i .
$A(s)$	Feasible action profiles in s : $A(s) \equiv \{(a_i, a_N) : a_i \in A_i(s), a_N \in A_N(s, a_i)\}$.
a_t	Period- t action profile: $a_t = (a_{i,t}, a_{N,t})$.
E	Edge set of the objective tree.
Γ	Objective state tree, $\Gamma \equiv (S, E)$.
f_t	Transition map (bijection): $f_t : \{(s_t, a_t) : s_t \in S_t, a_t \in A(s_t)\} \rightarrow S_{t+1}$, with $f_t(s_t, a_t) = s_{t+1}$.

Behaviors and induced distributions

σ_i	Agent behavior: maps each s to a distribution on $A_i(s)$.
σ_N	Nature behavior: maps each (s, a_i) to a distribution on $A_N(s, a_i)$.
Σ	Set of behavior profiles $\sigma = (\sigma_i, \sigma_N)$.
p_σ	Distribution on terminal histories induced by σ .

Table A.2. Agent A / A^+ : desires and evaluation

$\preceq$	Weak preference relation on terminal histories.
$\preceq$	Induced weak preference relation on behavior profiles.
d	Desire payoff function on terminal histories, $d : H_T \rightarrow \mathbb{R}$.
$\mathbb{E}(\sigma)$	Expected desire payoff under σ (risk-neutral evaluation with respect to d and p_σ).
$\Sigma_i^*(\sigma_N)$	Agent's most-preferred behaviors given Nature's behavior σ_N .

Table A.3. Agent B -family: credences and Bayesian learning

$\hat{\sigma}_N$	Agent's subjective model/credence about Nature's behavior.
$\bar{\Sigma}_N$	Finite set of deterministic contingency plans used in the Bayesian-learning representation of Nature.
$\bar{\sigma}_N$	A deterministic contingency plan for Nature.
$\hat{\sigma}_N(\cdot \mid h_0)$	Strictly positive initial prior on $\bar{\Sigma}_N$.
$\hat{\sigma}_N(\cdot \mid h_t)$	Posterior on $\bar{\Sigma}_N$ obtained by conditioning $\hat{\sigma}_N(\cdot \mid h_0)$ on the cylinder event determined by h_t .
$\iota(\bar{\sigma}_N, h_t)$	Indicator that $\bar{\sigma}_N$ matches Nature's realized moves along h_t .

Table A.4. Agent C-family: awareness, intervals, and costly deliberation*Tree operators*

$\text{Par}(s)$	Parent of state s in the objective tree, with $\text{Par}(s_0) = \emptyset$.
$\text{Chld}(s)$	Set of direct successor states of s .
$\text{Desc}(s)$	Set of descendant states of s in the objective tree; for terminal s , $\text{Desc}(s) = \emptyset$.
$\text{Depth}(s)$	Time index of state s , equivalently its distance from the root.

Intervals and scalar proxy values

$\underline{d}(s), \bar{d}(s)$	Lower and upper payoff bounds over terminal histories that pass through state s .
α	Interval-evaluation attitude used to convert an interval into a scalar proxy value.
$\hat{d}(s)$	Scalar proxy value assigned to an unexpanded state:

$$\hat{d}(s) = (1 - \alpha)\underline{d}(s) + \alpha\bar{d}(s).$$

Arrival awareness and deliberative-round objects

$\hat{M}_{s_t}^{\text{arr}}$	Arrival-awareness state set at s_t :
------------------------------	--

$$\hat{M}_{s_t}^{\text{arr}} := \{s_t\} \cup \text{Chld}(s_t).$$

k	Deliberative-round counter within the current deliberative episode. The baseline assessment after arrival awareness is $k = 0$.
M^k	State-awareness set available after k completed deliberative rounds in the current episode.
L^k	Leaf states of the current awareness tree M^k :

$$L^k := \{s \in M^k \mid \text{Chld}(s) \cap M^k = \emptyset\}.$$

L_{exp}^k	Expandable frontier states at round k :
--------------------	---

$$L_{\text{exp}}^k := \{s \in L^k \mid s \notin S_T\}.$$

s_ℓ	An indexed expandable frontier state in L_{exp}^k .
$s_\ell^{k,\star}$	Expandable frontier state selected for investigation at round k :

$$s_\ell^{k,\star} \in \arg \max_{s_\ell \in L_{\text{exp}}^k} B^k(s_\ell),$$

with ties resolved by the maintained tie-breaking convention.

k^*	Number of completed deliberative rounds when the deliberation procedure stops.
$\hat{M}_{s_t}^{\text{fin}}$	Final awareness tree after deliberation stops:

$$\hat{M}_{s_t}^{\text{fin}} := M^{k^*}.$$

Table A.5. Agent C^{Plan} : planning, active incoming plans, and off-plan review*Plan object* $\bar{\text{PL}}_{s_t}$ Outgoing plan formulated at reached nonterminal state s_t :

$$\bar{\text{PL}}_{s_t} \equiv (\bar{D}_{s_t}, \bar{\sigma}_{s_t}, \bar{\nu}_{s_t}).$$

 $\bar{D}_{s_t}$ Plan-domain state set of the outgoing plan formulated at s_t . It contains the within-plan states retained from the final awareness tree. $\bar{\sigma}_{s_t}$ State-contingent action prescription on the non-frontier states of the plan domain. At each planned non-frontier state s , $\bar{\sigma}_{s_t}(s) \in A_i^*(s)$. $\bar{\nu}_{s_t}$

Stored gross estimated continuation-value function on the plan domain. When the plan is formed from the present deliberative episode,

$$\bar{\nu}_{s_t}(s) := v^*(s) \quad (s \in \bar{D}_{s_t}).$$

These values do not subtract past deliberative-round costs.

Incoming and outgoing plans

Incoming plan

A plan $\bar{\text{PL}}_{s_{t'}}$, $t' < t$, available when the agent arrives at s_t .

Active incoming plan

An incoming plan usable at s_t : $s_t \in \bar{D}_{s_{t'}}$, $\bar{\sigma}_{s_{t'}}(s_t)$ is defined and belongs to $A_i(s_t)$, and $\bar{\nu}_{s_{t'}}(s_t)$ is defined.

Outgoing plan

The plan $\bar{\text{PL}}_{s_t}$ formulated at the end of deliberation at s_t . At a reached nonterminal decision state, Agent C^{Plan} leaves with an outgoing plan.

No active incoming plan

The arrival-side case in which no incoming plan supplies both a feasible current prescription and a stored gross estimated continuation value at s_t . The ordinary cases are the initial state before any plan has been formed and later arrival at a frontier state of the incoming plan. $\bar{P}_{s_t}$ Plan continuation at s_t when an incoming plan $\bar{\text{PL}}_{s_{t'}}$ is active:

$$\bar{P}_{s_t} := \bar{D}_{s_{t'}} \cap (\text{Desc}(s_t) \cup \{s_t\}).$$

This is the within-plan continuation retained from s_t onward, not the agent's full present awareness tree.*Planning when no incoming plan is active*

No-active-incoming-plan deliberation

If no incoming plan is active at s_t , Agent C^{Plan} deliberates exactly as Agent $C+$: she begins from $\hat{M}_{s_t}^{\text{arr}}$, runs the deliberation procedure of Section A.5.2, and forms an outgoing plan from the resulting final awareness tree.

Plan formation from final assessment

The outgoing plan retains only the on-plan continuation selected by the final assessment: at each discovered non-leaf state s reached by the construction, choose $\bar{\sigma}_{s_t}(s) \in A_i^*(s)$, close under Nature's feasible responses to that action, and store $\bar{\nu}_{s_t}(s) := v^*(s)$.

Plan frontier state

A state $s \in \bar{D}_{s_t} \cap \hat{L}_{s_t}^{\text{fin}}$. No planned action is stored at such a state, and no successor states beyond it are included in the plan. If the agent later reaches a nonterminal frontier state, the incoming plan is no longer active there.*Deliberation with an active incoming plan*